

RekenRangers: A configurable arithmetic learning tool designed for classroom use and educational research

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Preface

I chose this subject because I believe that building something that helps education is genuinely worth doing. The digital world we have entered offers enormous opportunities, but we still have much to learn about how to make the most of it, and there is real work to do before its power is turned toward the right ends. Tools that bring children, teachers, and researchers a little closer together feel like a good place to start. The fact that the project sat at the intersection of math, programming, and design only made it more enjoyable to spend a year on.

I want to thank my supervisor and promotor, Prof. Dr. Tom Schrijvers, for his guidance, his feedback, and the trust he placed in this project from the start. I am equally grateful to my co-supervisor, whose perspective shaped the direction of the research throughout. Special thanks go to Samuel Klumpers for his help deploying RekenRangers, and to Eline T'Joel and Kiara Willems for their advice, their time, and their patience in helping me understand the researcher's side of the problem.

Finally, I am grateful to my friends, my girlfriend, and my family for their constant support, their encouragement, and their willingness to listen to far more about arithmetic tools than they ever expected to.

Naten Kindekens

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Abstract

Digital tools play an increasingly important role in primary school education, and arithmetic learning is no exception. Two distinct categories of tools have emerged in this space. Classroom exercise platforms such as Automatus and RekenRangers offer adaptive practice, gamification, and child-friendly design that keep children engaged across daily use. Research tools such as OpenSesame and PsychoPy give educational psychologists the experimental control, data logging, and reproducibility they need to study how children actually learn arithmetic. The problem is that these two categories have evolved in isolation: classroom tools lack the configurability and data collection required for rigorous research, while research tools lack the motivational design and accessibility needed to keep children engaged across multiple sessions in a classroom setting. Researchers studying arithmetic learning are therefore forced to compromise on either scientific control or ecological validity, a compromise that directly affects the quality of the data their studies produce.

This thesis introduces RekenRangers (MathRangers), a web-based arithmetic learning platform built to bridge this gap. The tool combines a gamified, child-friendly environment built around a safari theme with the configurability needed for controlled experiments. Teachers and researchers can set up complex multi-block studies without writing any code, while children encounter the same tool as a regular practice environment with coins, puzzles, and animals to rescue. Twelve properties of a good arithmetic tool, six from the research side and six from the classroom side, are defined in the first part of the thesis and used as a structured framework for the design of the platform. RekenRangers was developed in close collaboration with co-supervisor Elien Bellon and with two master's students in pedagogical sciences, Eline T'joen and Kiara Willems, whose joint thesis project served as a concrete test case for the tool.

The platform was evaluated through its deployment in a real classroom study with 146 third-grade children across seven primary schools in Flanders, conducted over four training days in four experimental conditions. Due to time constraints on the analysis, the evaluation in this thesis is based on a subset of 107 of these participants. The evaluation combined data from the deployment, qualitative feedback from the two researchers who ran the study, and a short questionnaire completed by 45 of the participating children. The results show that RekenRangers can support a real multi-session classroom experiment from start to finish: more than 21,000 exercise records were collected and exported in structured form without preprocessing or data loss, and children remained visibly engaged. The gamification system

was used voluntarily by nearly all participants, with 98.5% of earned coins spent on puzzles and 99% of children completing at least one. The questionnaire results confirm that children experienced the tool as engaging and easy to use. The adaptive difficulty system produced meaningful differentiation between ability levels but did not converge on its target accuracy, indicating that the algorithm requires further calibration. The researcher feedback was positive, though both researchers were involved in the development of the tool, which limits how far that feedback generalises.

The thesis concludes that bridging the gap between classroom exercise tools and controlled research software is achievable in principle, while making the remaining trade-offs and open problems explicit. The most important limitations concern the flexibility of the tool, which cannot match the open-ended configurability of a programming environment, the adaptive algorithm, which needs further tuning, and several underdeveloped properties such as personalisation and feedback. These directions are discussed in the future work section. While RekenRangers still has parts that can be refined and extended, it is already a working tool that demonstrates the two worlds can be brought meaningfully closer together.

List of Abbreviations

Abbreviations

PISA	Programme for International Student Assessment
OECD	Organisation for Economic Co-operation and Development
GUI	Graphical User Interface
RQ	Research question
ZPD	Zone of proximal development
AI	Artificial Intelligence

Chapter 1

Introduction

Digital tools have become a standard part of education. From primary school to university, students interact with adaptive practice platforms, intelligent tutoring systems, learning management environments, and a growing number of AI-driven applications [20]. In primary school arithmetic alone, platforms such as Automatus [1] and Rekentuin [3] are used daily by hundreds of thousands of children across the Netherlands and Flanders. These tools have shown that **adaptive practice, gamification, and child-friendly design** can sustain engagement and support the development of arithmetic fluency.

In parallel, educational psychology researchers rely on a different category of digital tools to investigate *how* children learn arithmetic. Platforms such as OpenSesame [31] and PsychoPy [39] are designed to support experimental research, offering **precise control over stimulus presentation, complete data logging, and reproducible study configurations**. These tools have made it possible to study aspects of arithmetic development such as fact retrieval, strategy use, and metacognitive monitoring [6, 42].

The problem is that **these two categories of tools have evolved in isolation**. Classroom tools are built to engage children but are not built for research. Research tools are built for scientific control but are not built for classrooms. A researcher who wants to study arithmetic learning in a real classroom, often across a training period of several sessions rather than a single visit, is therefore forced to choose between two unsatisfactory options. The first is to use a research tool: in principle these tools can be extended with stories, rewards, and adaptivity, but doing so requires substantial programming, and in practice children are left with a sparse interface. The second is to use a classroom tool and accept that the experimental conditions cannot be controlled and the relevant data cannot be collected. Neither choice is good, and the compromise that results is rarely satisfactory.

This compromise is not just inconvenient. It directly affects the quality of the research that comes out of these studies. Arithmetic research with primary school children typically takes place over several sessions, spread across days or weeks, and depends on children staying motivated and focused throughout. When the tool used is unfamiliar and visually bare, three things tend to go wrong. First, children waste

time and mental energy figuring out how the tool works rather than focusing on the actual exercises. Second, sitting in front of a screen with no rewards, no story, and no sense of progress is exhausting, and children’s performance drops as the sessions go on [46, 26]. Third, the artificial setting of a stripped-down experiment looks nothing like the classroom the research is ultimately trying to understand, which limits how much the findings actually transfer back to real-world learning [12]. **A better tool, in short, means better research.**

This thesis introduces **RekenRangers (MathRangers)**, a web-based arithmetic learning platform designed to bridge the gap between classroom exercise tools and controlled research software. RekenRangers offers a gamified, child-friendly environment built around a safari theme, while also giving researchers fine-grained control over experimental conditions, complete data logging, and the ability to configure studies **without writing a single line of code. The goal is a single tool that children enjoy using and that researchers can rely on for their scientific work.**

RekenRangers was developed in close collaboration with the co-supervisor of this thesis, Elien Bellon, and with two master’s students in pedagogical sciences at KU Leuven, Eline T’joen and Kiara Willems. Their thesis project required exactly the kind of tool this work set out to build: a platform that could run a multi-condition arithmetic study in real classrooms while keeping children motivated across multiple training sessions. This collaboration shaped many of the design decisions and made it possible to evaluate the tool in practice, not just in theory.

To test whether RekenRangers actually delivers on its promise, the platform was deployed in a real classroom study with 146 third-grade children across seven primary schools in Flanders. Due to time constraints on the analysis, the evaluation presented in this thesis is based on a subset of 107 of these participants. Children worked with the tool over four training days, and afterwards filled in a short questionnaire about their experience. The two researchers who ran the study also provided written feedback on what it was like to use RekenRangers to set up, run, and monitor a real experiment. Together, the data, the questionnaire results, and the researcher feedback form the basis for the evaluation presented later in this thesis.

The results of this evaluation are encouraging, but with clear caveats. RekenRangers proved capable of supporting a real multi-session classroom experiment from start to finish: 21,522 exercise records were collected across four experimental conditions, the data was structured and exportable without preprocessing, and children remained visibly engaged with the tool throughout. The behavioural data shows that 98.5% of earned coins were spent voluntarily on puzzles and that 99% of participants completed at least one puzzle, both of which suggest that the gamification system was not just decoration but a genuine source of motivation. The questionnaire results support this picture, with children rating the tool positively on engagement and user friendliness.

At the same time, not everything worked as designed. The adaptive difficulty system differentiated between children of different ability levels but did not converge on its intended target accuracy, indicating that the algorithm needs further

calibration. Two related issues surfaced in the difficulty data: a ceiling effect, with several children reaching the maximum rating because the exercise pool did not extend far enough at the top end, and a plateau, with a sizeable group of children getting stuck around the boundary between two difficulty zones because the step in difficulty between those zones was too large. A submission bug led to a small number of duplicate records that had to be cleaned during analysis. And while the configuration of an experiment can be done without writing code, the interface for doing so still has room for improvement, particularly in helping researchers see and verify the experiment they are setting up.

Some properties of the tool are present but underdeveloped. Personalisation is functional but lacks the visual elements (avatars, personal dashboards) that would make the experience feel truly personal. The feedback system is minimal, providing only correctness information. And the flexibility of the tool, while sufficient for arithmetic studies that fit its built-in templates, cannot match the open-ended configurability of a programming environment. These are the natural directions for future work.

While there are remaining trade-offs and open problems, RekanRangers is a big step towards bridging the gap between classroom exercise tools and controlled research software. It provides a concrete design and implementation to analyze the fundamental problems that arise when aiming at this goal, and is a **working demonstration that the two worlds can be brought much closer to each other** than they currently are.

The remainder of this thesis is structured as follows. Chapter 2 reviews arithmetic research and existing research tools. Chapter 3 does the same for classroom exercise tools. Chapter 4 formalises the gap and defines the research questions. Chapter 5 introduces RekanRangers and its design. Chapter 6 describes the classroom deployment. Chapter 7 presents the evaluation results. Chapter 8 interprets these results and discusses the limitations. Chapter 9 answers the research questions and outlines future work.

Note on the use of generative AI

Generative AI tools were used in the preparation of this thesis. Their use was limited to assisting with writing, specifically the reformulation and stylistic refinement of text that had already been drafted by the author, and to support the search for relevant scientific literature. Any sources surfaced in this way were always retrieved, read, and critically assessed by the author before being cited, and no claim or reference was included on the basis of generative AI output alone. All scientific content, design decisions, analyses, and conclusions are the author's own. Generative AI was also used during the development of RekenRangers to assist with implementation tasks such as generating boilerplate code and debugging, as well as to create the visual assets used in the safari theme.

Chapter 2

Background and Motivation: The State of Arithmetic Research Tools

This chapter sets the stage for the rest of the thesis. It begins by establishing that arithmetic learning in primary school children is an active and relevant research field, then asks what kind of tools that research requires. Six properties of a good research tool are defined and used to evaluate two widely used platforms in the field: OpenSesame and PsychoPy. The chapter closes by examining what happens when these tools fall short, and why the consequences go beyond inconvenience for the researcher.

2.1 Arithmetic research in primary school children: an active field

Most people learn to count before they learn to read, and few skills follow a child further through life than the ability to do arithmetic. Yet despite its seemingly basic nature, how primary school children learn arithmetic remains one of the most actively studied topics in educational psychology. The reason for this is simple: we still do not fully understand how children learn it, why some struggle so much more than others, and what we can do about it.

Part of what makes this field relevant today is that the numbers coming out of large-scale assessments keep raising questions. Belgium’s average PISA mathematics score dropped from 508 in 2018 to 489 in 2022, the lowest since the country began participating [48], and the trend was already downward before the pandemic (Figure 2.1). The same pattern plays out internationally: as the OECD reports, “mean performance in mathematics across OECD countries fell by a record 15 points” between 2018 and 2022 [36]. Something is going wrong, and it is going wrong early.

The stakes are higher than just test scores. As McNeil et al. [33] put it: “High-quality mathematics education not only improves life outcomes for individuals but

2. BACKGROUND AND MOTIVATION: THE STATE OF ARITHMETIC RESEARCH TOOLS

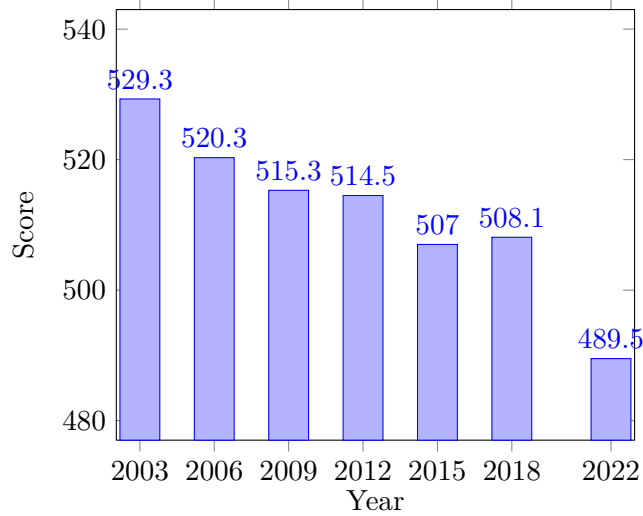


FIGURE 2.1: Evolution of Belgium’s PISA mathematics scores from 2003 to 2022, showing a clear downward trend in average performance over time. [48]

also drives innovation and progress across society”. They show that the arithmetic skills a child builds in primary school are not just useful for that year, but they form the foundation for everything that comes after [33]. Early numeracy skills are robust predictors of later mathematical achievement [33], with strong correlations observed between early assessments and standardized test scores in later years. Children who fall behind in arithmetic early tend to stay behind, and catching up becomes progressively harder as the curriculum moves on [33].

What makes this field particularly active is that arithmetic turns out to be far more complex than it looks from the outside. On the surface, solving a sum seems straightforward. But decades of research have made clear that children bring a whole collection of cognitive skills to the table when they do arithmetic [23]. These skills are still developing throughout primary school and that differ substantially from one child to the next. Some of these are specific to numbers: how well children process numerical quantities, or how quickly they retrieve arithmetic facts from memory [49]. Others are more general: how well they can hold information in mind [52], how flexibly they switch between strategies [25], and perhaps most interestingly, how accurately they judge whether their own answers are correct [42]. De Smedt [21] provides an overview of this landscape, noting that individual differences in mathematical performance are remarkably stable and already visible before children enter formal education. Understanding which factors drive these differences, and how they interact across development, remains an open and active research question.

Researchers at KU Leuven have been prominent contributors to this effort, particularly on the role of metacognitive monitoring¹ in arithmetic achievement [6, 7, 8] and on individual differences in numerical magnitude processing [49]. These are not

¹Metacognitive monitoring refers to a person’s subjective self-assessment of how well a cognitive task will be, is being, or has been performed [7].

isolated efforts. Similar questions are being pursued by research groups internationally, each approaching the problem from a different angle [23, 33, 52, 25].

All of this research depends on something practical: tools. Every study in this field depends on being able to observe children doing arithmetic under controlled conditions. Measuring not just whether they get the right answer, but how long it takes, how confident they are, and how their performance evolves over multiple sessions. The quality of that data depends directly on the quality of the tools used to collect it. As this thesis will argue, the tools currently available leave a significant gap between what researchers need and what is currently available. That gap is the starting point for this work.

2.2 What a good research tool needs

Before looking at what is currently available, it is worth being precise about what we are actually looking for. The previous section established that arithmetic research in primary school children is an active field with real scientific stakes. But good research questions can only produce good answers if the tools used to collect the data are up to the job. This section defines what that means in practice.

The six properties described below were developed through a combination of discussions with researchers in the field and a reading of how existing tools are evaluated in the literature on experimental software for educational psychology [38, 31]. They are not claimed to be exhaustive, but together they capture the core requirements that a tool must meet if it is to support rigorous², classroom-based research with children. Each property is introduced and briefly motivated here. The question of how well existing tools satisfy them is taken up in the following section.

Data Tracking The most basic requirement of any research tool is that it records what happens. In arithmetic research this means logging not just whether a child answered correctly, but also the response time, the exact answer given, the confidence rating if applicable, and the precise timestamp of each event. Without clean, structured, and complete data, no analysis is possible. This sounds obvious, but in practice many tools either do not log enough detail, make the export process too complex, or produce formats that require preprocessing before they can be used. A good research tool must make data collection automatic, complete, and straightforward to access.

Experimental control Research designs in educational psychology typically involve comparing groups under different conditions. For example, children who receive metacognitive prompts versus those who do not. To draw valid conclusions

²In the context of this thesis, “rigorous” research is understood in the sense of Shadish, Cook, and Campbell [45]: research whose design, data collection, and analysis are controlled tightly enough to support valid causal inferences. In practice, this means controlling extraneous variables, collecting complete and accurate data, and producing results that generalise beyond the specific experimental session.

2. BACKGROUND AND MOTIVATION: THE STATE OF ARITHMETIC RESEARCH TOOLS

from such a comparison, the researcher must be able to control precisely what each group sees and experiences. This includes controlling the type of exercises presented, the order in which they appear, which items are excluded, whether adaptive difficulty is active or not, and so on. Without this level of control, confounding variables (an outside, often hidden, third variable [44]) creep in and the internal validity of the study (the extent to which observed effects are caused by the manipulation rather than to other factors [14]) is compromised.

Reproducibility A finding is only scientifically meaningful if it can be replicated. For a tool to support reproducible research, it must be possible to save the complete configuration of an experiment and share it with other researchers so they can run the exact same study. This is closely tied to broader discussions in the open science community about transparency and replication [37].

Flexibility A research tool is rarely used for a single study. Experiments differ from each other in countless ways, and the same tool needs to support all of them without having to be rebuilt from scratch.

In practice, this means that the parameters a researcher might want to vary need to live outside the source code. This goes beyond simple settings like the number of trials or the timing of how long an item is shown. It also includes structural choices: what kind of stimuli appear (sums, comparisons, word problems, images), how those stimuli are presented and in what order, which prompts the child sees, what counts as feedback, and how a session is organised. If these aspects are hard-coded, the tool does not just become inconvenient to adapt. It becomes impossible for a researcher (or a teacher building their own experiment) to set up the study they actually want. Hard-coded choices turn into real limitations on the study design, which is why flexibility has to be designed in from the start.

Reliability Reliability is the property that is easiest to take for granted and most damaging to lose. A research tool is reliable when it behaves the same way for every participant, every session, every time. Timings are accurate (precision), inputs are captured without loss, the interface does not stutter at unpredictable moments (stability), and a crash during session three of a four-session study does not destroy the data from sessions one and two (robustness). None of this is optional. The moment a tool becomes unreliable, the data it produces becomes hard to interpret: “was that slow response time a real cognitive effect, or was the screen just slow to refresh that trial?” The concern is not theoretical, millisecond-scale timing errors in commonly used experimental equipment have been documented as a reason for replication failures in reaction-time research [40].

Reliability also means working everywhere the study is run, not just on the researcher’s development machine. Classroom research typically spans multiple sessions over days or weeks, on hardware the researcher does not control: the school’s laptops, the tablets in the library, a mix of browsers on a mix of operating systems.

A tool that performs well only under ideal conditions is, for practical purposes, unreliable.

Ease of setup A tool can do everything described above perfectly and still be unusable if setting it up takes three weeks and a degree in computer science. The researchers who drive the field are, with few exceptions, not software engineers. If the tool they are meant to use assumes otherwise, there are really only two outcomes: either studies get delayed while someone else writes the experiment, or the study gets reshaped to fit what is easy to implement rather than what the research question actually requires.

Ease of setup is more than the absence of programming. It is the whole experience of getting from “I want to run this study” to “the first child is doing the first exercise.” Can the researcher tell, by looking at the interface, what each setting does? Is the navigation between configuration screens clear enough that a teacher could manage it during a short training session? When something is misconfigured, does the tool flag the problem before a classroom full of children is already logged in? Visual clarity and guidance through the setup process are what make the difference between a tool a researcher can use independently and one that requires constant developer support.

2.3 Current research tools and their limitations

With a clear picture of what a good research tool needs to do, the next step is to look at what researchers actually have on their desks today. This section walks through the two tools that came up most often during discussions with the research team at KU Leuven: OpenSesame [31], a tool frequently used by my co-supervisor in research projects (including the master’s thesis discussed in Section 2.5), and PsychoPy [39], which is taught to psychology students. The discussion examines how each of them fares against the six properties introduced in Section 2.2. Both tools have been carefully designed and have earned their place in the field. The point of this section is not to argue that they are bad tools; it is to show that even the best tools currently available leave a specific kind of gap, and that this gap becomes critical the moment a study leaves the controlled lab and moves into a classroom.

The discussion starts with OpenSesame, then turns to PsychoPy, and ends by drawing out the limitations that both tools share. That final discussion is what motivates the shift in direction taken in Chapter 3.

2.3.1 OpenSesame

Following its authors, OpenSesame is described as “an open-source, graphical experiment builder for the social sciences” [31]. It was designed to make the construction of experiments in the social sciences more accessible than before by providing a graphical user interface (GUI) on top of a Python back-end. A researcher assembles an experiment by dragging items (trial loops, sketchpads, keyboard response components, inline scripts) into a tree structure that represents the flow of the study (as

2. BACKGROUND AND MOTIVATION: THE STATE OF ARITHMETIC RESEARCH TOOLS

shown in Figure 2.2). For anything the GUI does not cover, OpenSesame exposes an inline Python editor that lets the researcher drop directly into code [31]. It is cross-platform and free.

Against the six properties from Section 2.2, OpenSesame scores well on most of them. **Data tracking** is handled automatically: every trial produces a row in a structured log file, with response times, accuracy, and any custom variables the experimenter defines. **Experimental control** is one of its strongest points. The researcher has essentially full authority over what each participant sees, the order of stimuli, the exact timing of each event, and the conditions under which the experiment branches. **Reproducibility** is built into the format: an OpenSesame experiment is stored as a single file that bundles the configuration, the stimuli, and any inline scripts, which means that sharing the experiment with a collaborator is as simple as sharing that file. **Flexibility** is high because the inline Python escape hatch means there is almost no experimental design the tool cannot, in principle, express. **Reliability** is more nuanced. Timing precision was measured in an independent benchmark using a hardware Black Box Toolkit, where OpenSesame fell within a 1–5 ms range across visual, audio, and response measures, and performed consistently across Windows, macOS, and Ubuntu [10]. The broader, practical sense of reliability is harder to pin down. Because OpenSesame is a separate software package that has to be installed and configured correctly on each machine, we have heard of cases where it is difficult to get it running on classroom hardware. Overall, though, the tool can be considered reliable.

The property where OpenSesame runs into trouble is **ease of setup**. On the surface, the GUI looks approachable: drag some items, connect them, press run. But the moment the study steps outside the standard kind of experiment where a stimulus is shown and the participant gives a response, the researcher is pulled into Python. Adding a custom adaptive algorithm, implementing a non-trivial exclusion rule, conditionally showing feedback based on the previous answer, or linking trial sequences across multiple sessions: each of these require inline scripts. These scripts are not decoration, they become critical components of the experiment, and maintaining them demands real programming skill. A researcher without a background in Python is effectively stuck with what the GUI alone can do, which for any non-standard arithmetic study is not enough.

There is also a second layer to this problem that has little to do with programming. OpenSesame exposes a large number of configuration options across several panels and tabs, with defaults and vocabulary (loops, sketchpads, synchronisation back-ends) that only make sense once the underlying experiment-building model is already familiar. For an experienced user this is manageable, but for someone approaching the tool for the first time the learning curve is steep enough that they will typically need a trained research assistant in the room.

The OpenSesame team continues to develop the project (including a new generative-AI setup assistant [16] and a web version [30]), but the underlying picture is unchanged: anything beyond the standard GUI workflow still requires Python. OpenSesame is a powerful tool, but for researchers without a programming background it is limiting.

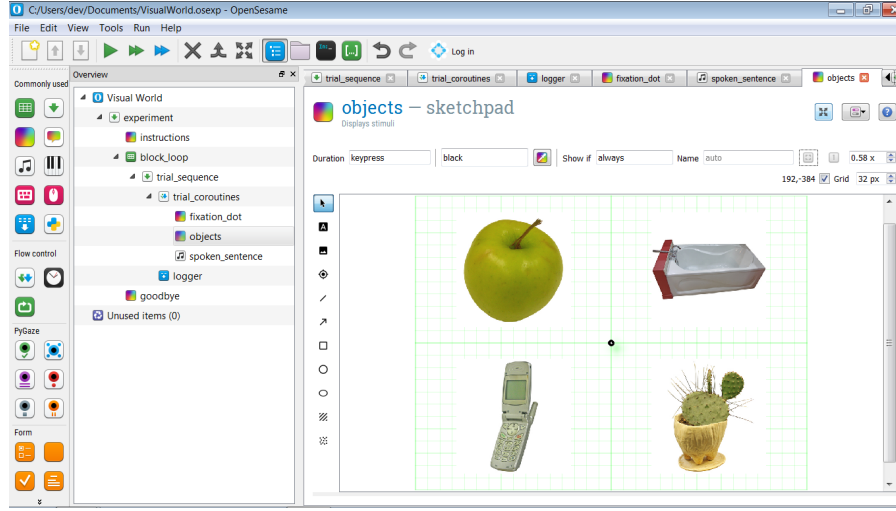


FIGURE 2.2: Screenshot of the OpenSesame graphical user interface, showing the experiment structure on the left and a sketchpad editor for designing visual stimuli on the right. [31]

2.3.2 PsychoPy

PsychoPy is the other open-source tool that comes up frequently in this work. Originally developed by Peirce as a Python library for psychophysics experiments [38], it has since grown into a full experiment-building environment with its own graphical Builder interface [39]. Where OpenSesame started from the GUI and added Python as an escape hatch, PsychoPy came from the opposite direction: it began as a pure coding library and later wrapped a graphical interface around it. This difference in origin matters because it affects how the tool behaves from a user’s perspective.

On the first five properties, PsychoPy performs comparably to OpenSesame. **Data tracking, experimental control, reproducibility, and flexibility** are all handled at a similar level: logging is automatic and extensible, the researcher has full control over what each participant sees, experiments can be saved and shared as structured files, and the underlying Python library means almost any design can be expressed in code. **Reliability** is where PsychoPy stands out the most. It was originally developed because existing tools did not offer the timing accuracy that psychophysics research demands [38], and independent benchmarks confirm that it remains among the most precise lab-based tools available, with mean visual, audio, and response timing under one millisecond [10]. It is also a mature project with a large active community of users and developers [39], which means bugs tend to be identified and fixed quickly. Like OpenSesame, however, it was designed for controlled lab environments rather than for the hardware mix of a primary school classroom.

That leaves **ease of setup**, where PsychoPy faces the same fundamental challenge as OpenSesame, but arguably in a more obvious form. The graphical Builder interface offers a drag-and-drop workflow that works well for simple experiments.

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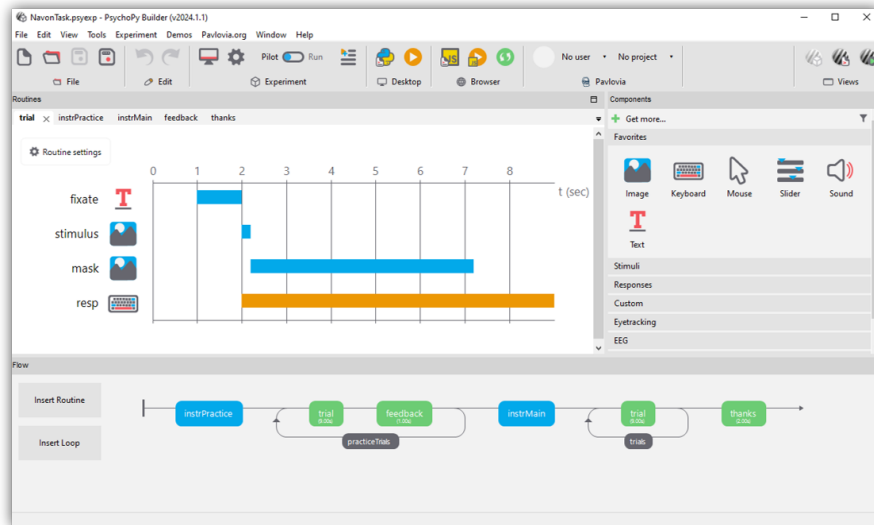


FIGURE 2.3: The PsychoPy Builder interface. Routines are displayed as timelines, and each Component maps onto a set of Python parameters. For experiments that exceed the Builder’s built-in Components, researchers insert Code Components containing raw Python, which effectively turns the study into a programming project.

PsychoPy’s own documentation is explicit about the trade-off: the Builder covers standard cases, and for anything more complex the researcher is expected to write raw Python code [39].

The interface itself reinforces this. A first-time user is confronted with terminology like Routines, Components, Backends, and Monitor configurations: concepts that make sense to someone with a programming background but are not self-evident to a researcher encountering the tool for the first time. The Builder GUI is genuinely well-designed and capable (Figure 2.3), but where OpenSesame attempts to hide the Python layer behind a familiar drag-and-drop tree, PsychoPy keeps it closer to the surface. The result is similar to OpenSesame but shifted further along the technical axis: an excellent tool for a researcher who already knows Python or is willing to learn it, but a steep entry point for anyone else.

At this point, a pattern is becoming visible. Both OpenSesame and PsychoPy satisfy the core research requirements but both run into the same wall when it comes to ease of setup. Figure 2.4 summarises this comparison. The six research-tool properties from Section 2.2 are listed as columns, and each tool is evaluated per property. The first five columns are green; the last one is not. This grid will be extended in Chapter 3 when the exercise-tool properties are introduced.

2.3.3 Shared limitations

The previous sections have shown that OpenSesame and PsychoPy are strong research tools. On five of the six properties defined in Section 2.2 they deliver exactly what a researcher needs: clean data tracking, full experimental control, reproducible

	Data tracking 	Experimental control 	Reproducibility 	Flexibility 	Reliability 	Ease of setup 
						
						

FIGURE 2.4: Comparison of OpenSesame and PsychoPy against the six research-tool properties from Section 2.2. Both tools satisfy all but the last.

setups, flexible designs, and reliability. But the picture changes completely when you step back and ask a different question: what is it actually like for a child to sit in front of one of these tools?

The answer, in most cases, is that it is not a pleasant experience. A typical arithmetic experiment built in OpenSesame or PsychoPy presents the child with white text on a black screen, one problem at a time, with no story, no colour, no reward, and no sense of progress (see example Figure 2.5). There is no character guiding them through the session or puzzle to unlock after ten correct answers. The experiment simply moves from one trial to the next until it is done. For a single short session in a quiet lab, this can work. But arithmetic research often takes place in classrooms, across multiple sessions spread over days or weeks, with children who are seven or eight years old. In that context, a black screen with numbers on it is not enough.

This is not an oversight by the developers of these tools. OpenSesame and PsychoPy were built to be general-purpose experiment builders, and they do that job well. The problem is the one identified in the previous sections: anything beyond the bare stimulus-response loop, gamification, a narrative wrapper, a child-friendly login, has to be built from scratch by the researcher, in Python. Because of the ease-of-setup wall, the researchers who need these features cannot build them. They know exactly what kind of experience they want the child to have, exercises that adapt to the child’s level, a colourful interface that feels like a game, metacognitive prompts wrapped in a story. But translating that vision into working code is a skill most of them do not have, and hiring or collaborating with a developer for every study is neither practical nor scalable.

This creates a frustrating situation. Researchers who study arithmetic learning in children know their population well. They know what kind of experience would actually serve the child sitting in front of the screen, and how that experience would in turn serve the data they collect. Yet the tools at their disposal make any of that difficult to deliver without significant programming effort. The result is a forced compromise: researchers know what they want, often something that resembles the classroom environment the research is trying to understand, but they cannot deliver

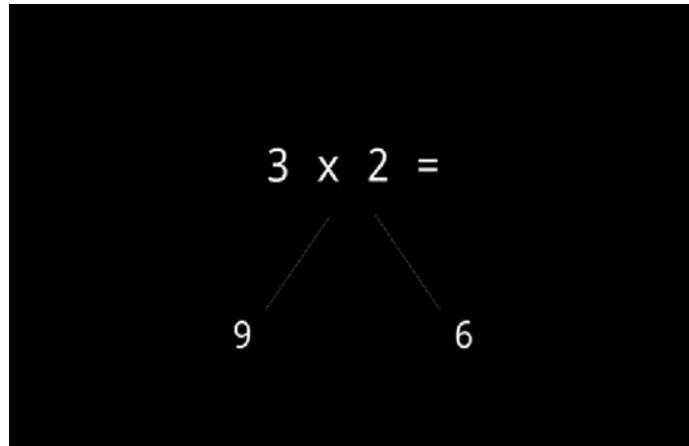


FIGURE 2.5: Example of a stimulus presentation screen created in OpenSesame, showing a simple arithmetic task with multiple-choice response options.

it. They end up running studies in conditions stripped of the context they actually care about.

Whether this matters, and how much it matters, is the question taken up in the next section.

2.4 Why this matters: the cost of a bad research tool

The limitations described in Section 2.3 are not just inconveniences for the researcher setting up a study. They have direct consequences for the quality of the research itself. When the tool a child interacts with during an experiment is unfamiliar, visually unappealing, and lacking in motivational features, the data that comes out of that experiment is affected in ways that are hard to control after the fact. This section discusses three mechanisms through which a poor research tool can compromise a study: reduced ecological validity, motivation drop across sessions, and confounding caused by tool unfamiliarity.

2.4.1 Ecological validity

Ecological validity refers to how well the conditions of a study match the real-world context the research aims to understand [12]. Bronfenbrenner argued that studies conducted in artificial laboratory settings, which differ from children’s everyday environments, often produce results that are difficult to generalise. Instead, he emphasised the importance of conducting research in natural settings where learning actually occurs, even if these settings are more complex [12].

Shadish, Cook, and Campbell make a similar point in the context of experimental design more broadly: highly controlled lab environments improve internal validity but at the cost of external validity, because the conditions no longer reflect the dynamic nature of real-world learning [45].

For arithmetic research with primary school children, this trade-off becomes very concrete. Children today learn arithmetic in environments that adapt to their level, use game-like elements, and let them exercise some control over what they do next. Adaptive difficulty, exercises that get easier or harder based on how the child is doing, keeps learners in a zone where the task is challenging enough to require effort but not so hard that they give up (i.e., zone of proximal development [ZPD]) [35]. Adaptive arithmetic learning apps have been shown to improve learning gains and engagement in kindergarten and early primary-school children [5]. Recent systematic reviews have shown that gamification can positively impact the academic performance, motivation, and engagement of primary school children [29]. A research tool that strips all of these features away does not just produce a less pleasant experience for the child, it produces an experience that no longer resembles the classroom context the research is ultimately trying to study. Whatever is measured under those conditions is, by Bronfenbrenner’s definition, less ecologically valid.

2.4.2 Motivation drop

The second cost is more practical and arguably more damaging: children lose motivation. Arithmetic research often spans multiple sessions across days or weeks, and this only works if children stay engaged throughout. Performance on cognitively demanding tasks declines as a function of time on task, and short breaks or changes in activity can partially restore it [46]. That finding is about fatigue within a single day, but it points to the same underlying issue at the multi-session scale: sustained mental effort produces measurable performance costs, and those costs accumulate when nothing in the task itself works against them.

When the tool itself offers no variation, no rewards, no story, and no sense of progress, the motivation problem is amplified. Research on the role of motivation in mental fatigue has shown that participants who receive motivational incentives are able to maintain performance levels on demanding tasks over extended periods, while those in non-rewarded conditions show clear performance declines [26]. In the context of arithmetic research with children, this translates to a concrete risk: if the experiment feels boring by session two, the data from sessions three and four may reflect disengagement rather than actual arithmetic ability. The researcher ends up measuring motivation, not mathematics.

The features that could prevent this, gamification, narrative framing, and immediate feedback, are known to support engagement and motivation over time [29]. They are also precisely the features that most research experiments lack and that the programming barrier described in Section 2.3 makes difficult to add.

2.4.3 The familiarity effect

The third mechanism is subtler but no less important. When a child encounters a completely unfamiliar tool during a research session, the tool itself becomes a variable in the experiment. The child has to figure out where to type an answer,

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what the buttons do, how the feedback works, and what is expected of them, all while supposedly demonstrating their arithmetic ability. A related effect is well documented in the testing literature: children tested by a familiar examiner score about 0.28 standard deviations higher than those tested by an unfamiliar one [22]. The same logic plausibly applies to the tool itself: unfamiliarity is a source of irrelevant cognitive load that lowers performance.

If the research tool were also used outside of the experiment, for regular classroom practice or homework, the tool would be invisible during the study in the way a pencil is invisible during a written test: present, but not a source of confusion. The data would more cleanly reflect what the study is actually trying to measure. There is a second advantage too: discoveries made through research, for example that metacognitive prompts improve arithmetic learning, could be fed back into the same tool and reach children as part of their regular practice. The tool would not only serve research but also benefit from it.

2.4.4 The bottom line

All three mechanisms point in the same direction. A research tool that is ecologically valid, motivationally engaging, and familiar to the children who use it is not a luxury or a nice-to-have. It is a prerequisite for producing data that accurately reflects what it claims to measure. In short: a better tool means better research. The question, then, is where to find such a tool. The natural place to look is at the platforms children already use for arithmetic practice in classrooms. That is the direction taken in Chapter 3.

2.5 A concrete example

The previous section argued, on theoretical and general empirical grounds, that a research tool offering little variation, reward, or sense of progress amplifies the motivation problem across multiple sessions. It is worth grounding that argument in a concrete case. This section looks at one recent master's thesis that ran a multi-session arithmetic training in OpenSesame, the kind of bare stimulus-response setup described in Section 2.3.3. The data from that study are limited, and the patterns described below are at the level of eyeballing rather than formal statistical testing. Still, they offer a concrete illustration of the motivation drop predicted in the previous section.³

Baeten [4] studied the relationship between metacognitive monitoring and the arithmetic learning process in 161 third-grade children. To produce a learning process, the children completed a four-day arithmetic training built in OpenSesame, solving the same set of two-digit multiplication exercises once per day across four consecutive school days. Each daily session consisted of 64 items, presented as white

³The study discussed here is an unpublished master's thesis [4]. Its results have not undergone peer review, and the patterns reported below should therefore be interpreted with appropriate caution. They are used here as an illustration of a mechanism, not as a definitive empirical claim.

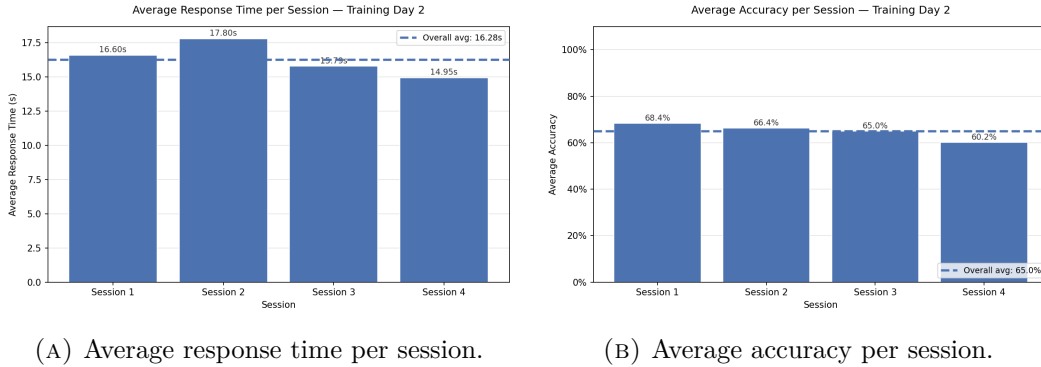


FIGURE 2.6: Within-day change across the four sessions of training day 2 in the arithmetic training of Baeten [4]. As the day progresses, children answer faster while becoming less accurate.

text on a black screen with feedback after every answer. This is exactly the type of multi-session, low-engagement experiment that Section 2.4 identified as being at risk of measuring motivation rather than mathematics.

The thesis itself notes this risk. In its discussion of limitations, the author observes that having children complete the same task on four consecutive days can reduce task interest, and that accuracy appeared to stagnate after the second training day rather than continuing to improve [4]. The thesis also already proposes a partial solution: it suggests that follow-up research could use rewards that depend on the participants' performance, so that children are sufficiently reinforced [4]. To examine the within-day pattern more closely, the data from the second training day were analysed session by session. Each training day was divided into four successive sessions of sixteen items, and average response time and average accuracy were computed for each session. The result is shown in Figure 2.6.

The two panels point in the same direction. After a small initial rise, the children's average response time falls steadily over the day, from 17.80 seconds in the second session down to 14.95 seconds in the fourth. If this were the only trend, it could be read as a sign of practice or growing fluency. But it is accompanied by the opposite movement in accuracy: the proportion of correct answers drops from 68.4% in the first session to 60.2% in the fourth. Children are not getting faster because they are getting better; they are getting faster while getting worse. That combination, shorter response times together with falling accuracy, is consistent with children rushing through the task: clicking quickly to reach the end rather than working through each problem. It appears to be the within-day counterpart of the motivation drop discussed in Section 2.4, and it shows up already on the second of four training days.

This reading fits what experienced arithmetic researchers report seeing in practice. Bellon (personal communication, 2025) describes three converging signals of waning engagement in this kind of multi-session study. The first is observational: testers notice changes in children's attitude and involvement, such as less task-

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focused behaviour and less sustained effort. The second is in the behavioural data itself, where disengagement leaves traces such as sudden drops in accuracy, exercises skipped by clicking through quickly, very short response times, and sometimes nonsense answers. The third concerns the learning trajectories: where one would ideally expect gradual progress across training days, that progress is often absent or only weak. The faster-but-less-accurate pattern in Figure 2.6 is broadly in line with these signals.

The lesson here is not that the study was poorly designed. The training was carefully constructed and ran on a capable, widely used research platform, and the thesis itself anticipated the motivation problem and proposed a reward-based fix. The point is rather that delivering such a fix is left entirely to the researcher. A research tool that provided engagement and performance-based rewards by default would lower that barrier, so that addressing the motivation drop does not depend on the individual researcher’s programming effort.

Chapter 3

Exercise Tools

The previous chapter established that the tools currently used for arithmetic research do their job as research instruments but fail as experiences for the children who use them. They lack the motivational, adaptive, and visual qualities that keep children engaged across multiple sessions, and the programming barrier makes it impractical for researchers to add these qualities themselves. The question that follows naturally is: where do these qualities already exist?

The answer is not far away. Across primary schools in Belgium and the Netherlands, children already use digital tools for arithmetic practice on a daily basis. Platforms like Automatus [1] and Rekentuin [3] are designed from the ground up to be engaging, adaptive, and easy for children to use. They look like games, they feel like games, and children are willing to spend time on them without being asked twice. In other words, these tools already solve many of the problems identified in Section 2.4: they offer the ecological validity, the sustained motivation, and the familiarity that research experiments lack.

This chapter examines exercise tools as a possible solution direction. It begins by defining what makes a good exercise tool, then walks through two prominent examples in the field, and ends by asking the inevitable follow-up question: if exercise tools solve the engagement problem, can they also serve as research tools? The answer, as will become clear, is not quite.

3.1 What a good exercise tool needs

Where Section 2.2 asked what a research tool needs, this section asks the same question for the other side: what makes a good tool for arithmetic practice in the classroom? The properties listed below were developed through discussions with primary school teachers, educational researchers, and the co-supervisor of this thesis. They are not claimed to be exhaustive, but together they capture what practitioners in the field consistently identify as important. Where possible, connections to the educational psychology literature are drawn to ground these observations in theory.

Adaptivity Perhaps the single most important property of a good exercise tool is that it adapts to the child using it. Not every child in a classroom is at the same level, and a tool that presents the same exercises to everyone will inevitably bore the strongest students and frustrate the weakest ones.

This idea has deep theoretical roots: Vygotsky’s zone of proximal development (ZPD) is the gap between what a child can do on their own and what they can do with a little help [50]. Csikszentmihalyi’s flow theory makes a closely related point about engagement more generally, namely that people concentrate best when the difficulty of a task is well-matched to their skill level [18]. In the specific case of arithmetic learning apps for young children, this is more than theory: Bang, Li, and Flynn showed that an adaptive arithmetic app produced significantly larger learning gains and stronger engagement than a non-adaptive control [5]. An exercise tool that does not adapt is, in effect, guaranteeing that most of the children using it will be working below or above their level most of the time.

Engagement Adaptivity keeps the difficulty right, but it does not on its own make a child want to open the tool again tomorrow. Engagement is about everything else that makes the experience feel worthwhile: a story that gives context to the exercises, visual rewards for progress, puzzles to unlock, a character to interact with. These are gamification elements, and systematic reviews of gamification in education consistently find that elements such as points, levels, narrative framing, and immediate feedback improve motivation and engagement in mathematics [29]. For an exercise tool aimed at primary school children, engagement is not optional. A tool that children do not want to use is a tool that does not get used, regardless of how well-designed its exercises are.

Feedback Children need to know whether they got the answer right, and ideally they need to know it immediately. Delayed or absent feedback leaves the child uncertain about whether they are learning, and it removes one of the most natural drivers of improvement: the desire to correct a mistake you just made. Meta-analyses of educational feedback confirm that corrective, computer-delivered feedback is among the most effective forms, particularly for new skills [51]. Interestingly, more is not always better: in a recent study with elementary children, feedback that simply showed the correct answer led to greater persistence and more flexible strategy use than feedback that also added an explicit “right/wrong” verification cue [34]. The key is that feedback is timely, clear, and meaningful.

Personalisation Personalisation gives the child the feeling that the tool is theirs. This has two sides. The first is functional: letting children choose which type of exercise to practise, whether they want to see a timer, or what difficulty level they prefer. This kind of choice can give children a sense of ownership: a child who decides to practise multiplication because they know it is their weakest area is doing something more valuable than just answering another question. The second side is personal identity: an avatar, a nickname, a progress tracker that shows their journey,

a customised home screen. These elements may seem cosmetic, but they transform a generic program into something that feels like it belongs to the child, which in turn makes the child more likely to come back to it [17].

Accessibility An exercise tool is only useful if children can actually reach it. In practice this means the tool should run in a web browser, work on the devices that schools actually have (laptops, tablets, Chromebooks), and not require any software installation. It also means the tool should be available outside the classroom, so that children can practise at home if they want to.

User friendliness Finally, the tool has to be usable by its target audience, which in this case is children between roughly seven and eleven years old. This means large buttons, clear icons, simple navigation, and a visual language that children understand without needing instructions. Every unnecessary interface element adds extraneous cognitive load, the kind of mental effort that does not contribute to learning, and so takes working memory away from the actual exercise [32].

3.2 Existing exercise tools: Automatus and Rekentuin

3.2.1 Automatus

Automatus [1] is a web-based arithmetic practice platform used in primary schools across Flanders and the Netherlands. It was designed with a clear purpose: to help children automate their arithmetic skills through repeated, structured practice. Teachers assemble customised “lands” consisting out of tiles, where each tile corresponds to a specific type of problem (for example, additions up to 20, or the multiplication table of 7). Children then work their way through these tiles, with the map giving each session a visible sense of direction and progression. Figure 3.1 shows the tile screen children encounter while practising.

On the six exercise-tool properties from Section 3.1, Automatus scores well on five of them. **Adaptivity** is handled in a specific way: rather than adjusting the difficulty of the exercises, Automatus adapts the goal time the child has to answer them. This is what the platform calls *adaptive automatisisation* as opposed to *adaptive practice*, and it positions itself as one of the first platforms built around this distinction¹. **Engagement** is supported by the map-and-tile structure, which gives children a visible sense of progress, and by response-time tracking that lets them compare their current attempt against their personal best, adding a layer of self-competition. **Feedback** is well-handled: when a child answers incorrectly, the exercise progress pauses and the platform shows a short explanation of the correct approach. **Accessibility** is excellent: Automatus runs in any browser and is also available as a tablet application, so it works on the devices schools and families actually have. **User friendliness** is equally strong: the interface is clean and intuitive enough that children can navigate it without adult help.

¹<https://automatus.be/scholen/faq/wat-is-adaptief-automatiseren/>



FIGURE 3.1: Screenshot of the Automatus interface from the pupil’s perspective, showing the tile-based landscape through which learners navigate exercises.

The one property where Automatus does not fully deliver is **personalisation**. The functional side is partially present: children can in some cases choose which exercises they want to work on, and the performance tracking gives a sense of individual progress. But the identity side is missing: there is no avatar, no personal profile beyond a name, no visual element that makes the tool feel like it belongs to the child rather than to the classroom. This is a minor gap in an otherwise very well-designed tool.

3.2.2 Rekentuin

Rekentuin (Math Garden) is a web-based arithmetic learning environment developed at the University of Amsterdam by Straatemeier, Klinkenberg, and van der Maas [47]. From the start it was built with a dual purpose: serving children as a practice platform and serving researchers as a scientific instrument for studying arithmetic development [47]. It has grown into one of the most widely used arithmetic tools in the Netherlands, with more than 400,000 primary school children practising on it [11], and is now maintained and distributed as part of the Prowise Learn [2].

The platform organises its content around a world map with multiple thematic islands, each representing a different mathematical domain. Children navigate between islands and work through exercises that range from basic addition to fractions and percentages. Figure 3.2 shows the main interface.

On the six properties, Rekentuin delivers strongly across the board. **Adaptivity** is the cornerstone of the platform’s design. Rekentuin’s adaptive engine combines the Elo rating system with an item response model, continuously estimating how able each child is and how hard each item is, and then choosing items so that the child answers correctly with a probability of roughly 75% [27]. Children can also choose between three difficulty levels within each game, adding a layer of self-regulation on top of the algorithm. **Engagement** is sustained through the island-based world,



FIGURE 3.2: Screenshot of the Rekentuin interface from the pupil’s perspective, showing the different continents from which learners can select different exercises.

a variety of games across different mathematical topics, and a coin system: faster and more accurate answers earn more coins. The platform also highlights recommended games and “super games” based on the child’s performance, gently steering practice toward areas that need attention. **Feedback** is immediate. Children see whether their answer was correct after each problem, and the coin reward provides a secondary signal of how well they performed relative to the time limit. **Personalisation** is where Rekentuin distinguishes itself most clearly from Automatus. The coins children earn are tied to an avatar that they can dress, decorate, and customise, giving every child a visible, personal identity within the platform. Combined with the personalised game recommendations and the self-selected difficulty levels, the experience feels suited to each individual. **Accessibility** is strong: Rekentuin runs in any web browser, including on mobile devices, and data from Klinkenberg et al. show that children completed roughly a third of their exercises outside school hours [27], indicating that the platform is genuinely used at home as well as in the classroom. **User friendliness** is well-designed for the target age group, with large interactive elements, colourful visuals, and a navigation structure built around the world and islands that children understand intuitively.

At this point, a similar comparison to the one drawn for OpenSesame and PsychoPy in Section 2.3 is useful. Figure 3.3 evaluates both Automatus and Rekentuin against the six exercise-tool properties. Automatus falls just short on personalisation, while Rekentuin delivers on all six. These tools do exactly what they were

	Adaptivity 	Engagement 	Feedback 	Personalisation 	Accessibility 	User friendliness 
						
						

FIGURE 3.3: Automatus and Rekentuin evaluated against the six exercise-tool properties from Section 3.1. Both perform well across the board; Automatus falls slightly short on personalisation.

designed to do. The question that follows is whether they can also serve the needs of researchers. That is the subject of the next section.

3.3 Why exercise tools alone are not the answer

The previous section showed that Automatus and Rekentuin are excellent exercise tools. They are adaptive, engaging, well-designed, and genuinely enjoyed by children. A natural question follows: if these tools already solve the engagement and motivation problems that plague research tools, could they not simply be used for research as well?

To answer this, the same six research-tool properties from Section 2.2 can be applied to both platforms. The result, summarised alongside the exercise-tool evaluation in Figure 3.4, shows a clear pattern. On three of the six research properties, both tools perform well. On the other three, they fall short in ways that matter.

The three properties that hold up are straightforward. **Reproducibility** is well-supported: once a teacher has configured a set of exercises for a class, that configuration is saved and can be reused next semester without effort. **Reliability** is equally strong. Both Automatus and Rekentuin are mature, commercially maintained products used daily by hundreds of thousands of children. They are stable, they do not crash mid-session, and they behave consistently across devices. **Ease of setup** is, unsurprisingly, a strength rather than a weakness. Both tools were built for teachers, not programmers. Setting up a class, assigning exercises, and managing student progress is done entirely through a graphical dashboard with no code involved.

The problems emerge in the remaining three properties. **Data tracking** is the first issue. Both platforms collect data on student performance, and teachers can see summaries of accuracy and progress over time. But the level of detail is not what a researcher needs. Arithmetic research requires the exact response time for every individual trial, the precise answer the child gave (not just whether it was correct), and ideally the timestamp of every screen transition. These fine-grained

3.3. Why exercise tools alone are not the answer

	Data tracking	Experimental control	Reproducibility	Flexibility	Reliability	Ease of setup	Adaptivity	Engagement	Feedback	Personalisation	Accessibility	User friendliness
Automatus	✗	✗	✓	✗	✓	✓	✓	✓	✓	✗	✓	✓
rekentuin	✗	✗	✓	✗	✓	✓	✓	✓	✓	✓	✓	✓

FIGURE 3.4: Combined comparison of Automatus and Rekentuin against both the exercise-tool properties (Section 3.1) and the research-tool properties (Section 2.2). Both tools excel on the exercise side but fall short on data tracking, experimental control, and flexibility.

data points are what allow researchers to distinguish, for example, between a child who retrieved a fact from memory in under two seconds and one who calculated the answer in five. Exercise tools log enough to inform a teacher; they do not always log enough to inform a study.

Experimental control is the second and arguably most fundamental problem. As discussed in Section 2.2, controlled experiments depend on the researcher being able to dictate what each participant sees and under what conditions. Exercise tools are designed around a partly opposite principle: they are built to give children autonomy, with the system adapting on its own and the teacher setting broad goals rather than trial-by-trial sequences. Some features can be configured by the teacher, and it is not impossible to get two children to see the same set of exercises. But for the kind of fine-grained control a research design typically needs (disabling specific features for a single group, forcing a fixed item sequence, switching off adaptivity while keeping everything else the same) the platforms simply were not designed with this in mind, and the level of control an OpenSesame-style tool offers is out of reach.

Flexibility is the third gap, and it is closely tied to the previous one. As also discussed in Section 2.2, flexibility in a research context is the ability to set up a wide range of experimental designs without rebuilding the tool. The configuration options exercise tools expose are designed for pedagogical goals, not experimental ones. A researcher who wants to swap the adaptive algorithm for a different one, introduce a new kind of task, or restructure the session into multi-block sequences with custom rules between blocks has no way to express any of that within the platforms. The kind of flexibility a programming language offers is absent, and the graphical interface itself was never built to support the structured designs that some research studies require.

Figure 3.4 brings both evaluations together. The research-tool properties are on the left, the exercise-tool properties on the right. Automatus and Rekentuin fill the right side almost entirely with green. The left side tells a different story: reproducibility, reliability, and ease of setup are covered, but data tracking, experimental control, and flexibility are not. Exercise tools, on their own, are not the answer. But the qualities they bring to the table are exactly the qualities that Chapter 2 identified as missing from research tools. The next section brings all four tools and all twelve properties together to make this gap fully visible.

	Data tracking	Experimental control	Reproducibility	Flexibility	Reliability	Ease of setup	Adaptivity	Engagement	Feedback	Personalisation	Accessibility	User friendliness
Automatus	✗	✗	✓	✗	✓	✓	✓	✓	✓	✗	✓	✓
rekentuin	✗	✗	✓	✗	✓	✓	✓	✓	✓	✓	✓	✓
OpenSesame	✓	✓	✓	✓	✓	✗	✗	✗	✗	✗	✗	✗
PsychoPy	✓	✓	✓	✓	✓	✗	✗	✗	✗	✗	✗	✗

FIGURE 3.5: Full comparison of all four tools against the twelve properties defined in Sections 2.2 and 3.1. Exercise tools (top) score well on exercise properties but lack key research properties. Research tools (bottom) score well on research properties but offer none of the exercise-tool qualities.

3.4 The gap: no existing tool covers both sides

The previous sections built up the comparison piece by piece. Section 2.3 evaluated OpenSesame and PsychoPy on the research-tool properties, Section 3.2 evaluated Automatus and Rekentuin on the exercise-tool properties, and Section 3.3 tested whether exercise tools could also serve as research tools and found that they could not. The one combination left to make explicit is the reverse: how do the research tools score on the exercise-tool properties?

Section 2.3.3 answered this question. OpenSesame and PsychoPy were not built to be exercise tools, and they offer none of the six properties from Section 3.1 out of the box. There is no built-in adaptivity, no gamification, no avatar system, no child-friendly login. All of these could in theory be added, but only by writing them from scratch in Python, which brings the researcher straight back to the same programming wall that already made ease of setup the weakest point on the research side.

Figure 3.5 brings everything together. All four tools are evaluated against all twelve properties: the six research-tool properties on the left, the six exercise-tool properties on the right. The pattern is now fully visible. Automatus and Rekentuin fill the exercise columns with green but leave gaps in the research columns. OpenSesame and PsychoPy do the reverse. Each tool excels at what it was designed for. None covers both sides.

This is not a criticism of any of these tools individually. Each was built for a specific purpose and serves that purpose well. But for a researcher who wants to run a controlled experiment in a classroom, with children who stay motivated across multiple sessions, using a tool that feels familiar rather than foreign, there is currently no option that works. That gap is formalised in the next chapter.

Chapter 4

Problem Statement

Chapters 2 and 3 approached the same gap from two directions. Chapter 2 showed that the research tools available today offer strong scientific foundations but create experiences that are unsuitable for children in a classroom. Chapter 3 showed that the exercise tools children already use deliver exactly the kind of engaging, adaptive, child-friendly experience that research tools lack, but fall short on the data tracking, experimental control, and flexibility that researchers need.

Figure 3.5 made this visible. The grid splits cleanly in two, and no existing tool fills both halves. Researchers who want to run a controlled arithmetic study in a classroom are forced to choose: either a research tool with no motivation, no story, and no adaptivity for the child, or a classroom tool that does not give them the control, the data, or the structure their research design needs.

Problem

There is currently no digital arithmetic tool that combines the engagement, adaptivity, and child-friendly design of classroom exercise tools with the data tracking, experimental control, and flexibility of controlled research software. Researchers studying arithmetic development in children are forced to compromise on either scientific rigour or ecological validity, which directly affects the quality of the data their studies produce.

Goal and Research Questions

Design, develop, and evaluate a digital arithmetic learning platform that satisfies both the exercise-tool properties and the research-tool properties identified in this thesis. The platform should combine a gamified, child-friendly environment with the flexibility needed for controlled experiments, and it should be operable by teachers and researchers through a graphical interface without any programming knowledge. It should be usable both as a regular classroom exercise tool and as a research instrument, so that children can become familiar with the platform before participating in a study, and so that findings from research can flow back into classroom practice.

This goal is broken into three research questions:

- RQ1** Can a single digital platform satisfy both the exercise-tool properties (adaptivity, engagement, feedback, personalisation, accessibility, user friendliness) and the research-tool properties (data tracking, experimental control, reproducibility, flexibility, reliability, ease of setup)?
- RQ2** Can the platform support a real multi-session classroom experiment with over 100 participants across multiple conditions, producing complete and reliable data, keeping children engaged, and running an adaptive difficulty system that performs as designed?
- RQ3** How do children evaluate the engagement and user friendliness of the platform, and how do teachers and researchers evaluate the ease of setup?

RQ1 is a design question: it asks whether the twelve properties can coexist in a single tool, and it is answered through the design and implementation described in Chapter 5. RQ2 is an empirical question: it tests whether the tool actually works as a research instrument by examining its performance during a real classroom study, including the completeness and quality of the collected data, the behaviour of the adaptive system, and the sustained engagement of the participants across sessions. RQ3 is a qualitative question that captures the user experience from two perspectives: the children, through a short questionnaire, and the teachers and researchers, through structured feedback. Together, these three questions cover the full scope of the thesis. The experimental setup for RQ2 and RQ3 is described in Chapter 6, and the results are presented in Chapter 7.

Chapter 5

RekenRangers

The previous chapters identified a gap between classroom exercise tools and controlled research software, and Chapter 4 formalised that gap as a problem statement. This chapter describes RekenRangers¹, the platform built to address it. RekenRangers is a web-based arithmetic learning tool designed for primary school children aged seven to eleven. It is built to function both as a daily classroom exercise tool and as a configurable research instrument, so that the same platform children use to practise arithmetic can also serve as the environment in which researchers run controlled experiments.

The platform is hosted on a KU Leuven server and accessible through any modern web browser at <https://rekenrangers.cs.kuleuven.be/>, which means it runs on the laptops, tablets², and Chromebooks that schools already have without requiring any installation. The back-end is written in Python using the Flask web framework [24], with a MySQL database for persistent storage. The front-end uses HTML, CSS, and JavaScript, with the Bootstrap library for responsive layout. The full codebase is maintained in a private GitHub repository.³

This chapter begins by describing the design process and the decisions that shaped the platform at a high level, then walks through the tool’s approach to each of the twelve properties identified in Chapters 2 and 3.

5.1 Design process and guiding principles

RekenRangers was not designed in isolation. Building a tool that needs to work for children, teachers, and researchers at the same time is hard to do without input from all three groups, and the design process was set up to make at least some of that input possible. While the project did not involve frequent direct meetings with

¹Although the application is officially named *RekenRangers*, the English name *MathRangers* may also be used throughout this thesis for readability and convenience.

²RekenRangers runs in any tablet browser, but the interface has not been specifically optimised for tablets, and there is currently no tablet application. Both are possible directions for future work.

³The author of this thesis intends to set an open-source license on the repository in the future, so that the platform can be reused, extended, and inspected by other researchers.

teachers or children, advice was sought from these groups at key moments, and the design was continuously discussed with the researchers who would eventually use the tool in their own studies. Research in educational technology has consistently shown that involving stakeholders early and continuously in the design process leads to tools that are more likely to be adopted and that better fit the needs of the people who will actually use them [19].

The core development, both back-end and front-end, was carried out by the author of this thesis. Design decisions, however, were shaped through continuous collaboration with several key partners. Throughout the project, the author's supervisor, Prof. Dr. Tom Schrijvers, provided guidance on the overall direction of the work and feedback on its development. On the educational-research side, the co-supervisor of this thesis, Elien Bellon, contributed both the scientific grounding for the platform's features and a concrete benchmark study to replicate, drawing on her own research on metacognitive monitoring in arithmetic [6, 7, 8]. Two master's students in pedagogical sciences at KU Leuven, Eline T'joen and Kiara Willems, were also closely involved, since their own theses depended on using RekenRangers in classroom settings. Their involvement meant that design decisions were tested against real use cases from the start: if a feature did not make sense to someone planning a study with third-graders, it was reworked. A final, less direct source of input was the set of existing tools discussed in Chapters 2 and 3. These tools served a dual role. RekenRangers and Automatus provided models for what a child-friendly arithmetic tool looks and feels like, from adaptive difficulty to avatar systems to coin-based reward structures. OpenSesame and PsychoPy provided a clear reference for the kinds of experimental control and data logging a research tool must offer, but they also provided a vivid image of what happens when engagement is not a priority. The black-screen, stimulus-response experience that characterises most experiments built in these tools (see Figure 2.5) was a constant reminder of what RekenRangers needed to avoid.

This combination of perspectives shaped two core principles that guided the design throughout.

The first was that the replication of the Bellon et al. study [6] had to be possible. This study, which investigates the relationship between metacognitive monitoring and arithmetic performance in children, lived in the back of the author's mind throughout development: it was a concrete example of a study that had already been run in OpenSesame and that the researchers involved had found difficult to set up there. It was not used as a checklist that every feature was tested against, but it acted as a recurring reference point that kept the development grounded in a real research need rather than in abstract feature lists.

The second was a matter of audience. RekenRangers needed to be configurable by people who do not write code and feel like a game to the children using it. These are essentially restatements of the central goals identified in Chapters 2 and 3, but they were treated as design principles in their own right: every screen, every flow, and every configuration option was evaluated against whether it served a non-programming researcher on one side and a seven-year-old on the other.

Generative AI tools were used during development to assist with specific imple-

mentation tasks such as generating boilerplate code and debugging. All architectural decisions, feature design, and user interface design were made by the author in collaboration with the partners described above.

The following section describes how each of the twelve properties from Chapters 2 and 3 was addressed in the design and implementation of RekenRangers.

5.2 Approach: design decisions per property

This section walks through each of the twelve properties from Chapters 2 and 3 and describes how RekenRangers addresses them. The research-tool properties are discussed first, followed by the exercise-tool properties.

5.2.1 Data tracking

Section 2.2 defined data tracking as the most basic requirement of any research tool: the tool must record what happens, completely and automatically. In RekenRangers, every exercise completed by every child produces a full data record that is stored in the database. The level of detail was determined in close collaboration with Eline T’joen, Kiara Willems, and co-supervisor Elien Bellon, to ensure that the logged data would be sufficient for the kinds of analyses that arithmetic research requires.

Each record captures the full context of a single exercise: who answered, the operation and operands presented, the answer given, whether it was correct, and the response time. The record also includes the difficulty level, whether the timer was visible, the child’s current Elo rating and the change in rating produced by that exercise, and the difficulty zone the child was in at the time. When the metacognitive self-assessment is enabled, the confidence judgement is logged together with a calibration field capturing whether the self-assessment matched the actual outcome. Response times are recorded separately for the exercise, the confidence judgement, and the feedback screen, which lets researchers analyse not just what children answered but how long they spent on each phase of the trial. The complete list of fields is given in Appendix B.

This data is accessible in two ways. The teacher dashboard, when a student is selected from the group overview, shows the full list of exercises that student has completed, at a level of detail aimed at the teacher rather than at the researcher. For research purposes, the group screen offers an export function that generates a single CSV file containing every exercise completed by every student in the group. This file can be opened directly in Excel or imported into statistical software for analysis. The export is designed to require no preprocessing: each row is one exercise, each column is one variable, and no manual cleaning or restructuring is needed before the data can be used.

5.2.2 Experimental control

Experimental control is the property that most clearly separates a research tool from a classroom tool. Section 2.2 defined it as the ability to dictate precisely what each

participant sees and under what conditions, so that the researcher can isolate the variables they are studying. In RekenRangers, experimental control is implemented through a layered settings system in the teacher dashboard, where each group of children can be configured independently.

The first layer covers the general exercise settings. For any group, the teacher or researcher can choose which arithmetic operations are available, how difficulty is determined (adaptive, fixed, or chosen by the child), whether a timer is shown, and whether metacognitive self-assessment prompts are enabled. These settings are sufficient for classroom use and for simple studies where the main concern is controlling what type of exercises children see. The full list of configurable settings is provided in Appendix C.

The second layer covers the research settings (Figure 5.1), where the experimental control becomes precise. The most important option here is *research mode*. When research mode is enabled, the interface changes in ways that are invisible to the child but critical for the researcher. Mouse interaction is disabled entirely, so that all input happens through the keyboard. Buttons and interface elements that are not essential to the exercise flow are hidden or deactivated, reducing the chance that a child accidentally navigates away from the task or gets distracted by something outside the experiment. Several pieces of interface text are also adjusted: in normal mode some labels offer the child additional options that are no longer available in research mode, so the corresponding text is replaced with a research-mode equivalent. In short, research mode strips the interface down to what the experiment requires and nothing more.

Within the research settings, a set of item generation rules gives the researcher fine-grained control over which exercises can appear. Identical exercises can be blocked from appearing twice within the same session. Tie problems, where both operands are the same number (such as $4 + 4$ or 7×7), can be excluded because they are known to be solved through different processes than non-tie problems [15]. Exercises that produce the same result can be prevented from appearing consecutively. Commutativity can be enforced, so that 3×7 and 7×3 are registered as the same exercise. The position of the larger operand can be balanced across trials to prevent children from developing a position bias. Beyond these structural rules, the researcher can define exclusion lists for specific operand values and results, for example excluding 0, 1, or 10. A fixation screen can also be enabled before each exercise, which is standard practice in reaction-time research to ensure the child is focused before the stimulus appears.

Effort also went into ensuring that children cannot inadvertently break the experimental flow. In research mode, there is no way to skip an exercise, navigate to the puzzle screen mid-session, or exit the exercise sequence (except through browser actions). Answers are constrained to valid numerical input, and the interface enforces a fixed progression from exercise to self-assessment to feedback to the next exercise. In a classroom where twenty children work at once and no researcher can watch every screen, getting these things right is what keeps a session’s data usable.

The combination of the general settings and the research-specific controls means that a researcher can configure two groups within the same classroom to see identical

5.2. Approach: design decisions per property

Onderzoekinstellingen

Onderzoeksmodus
Schakel deze modus in wanneer leerlingen deelnemen aan het onderzoek. De interface wordt dan aangepast om afleiding te beperken en de afname zo gecontroleerd mogelijk te laten verlopen. In deze modus worden onder andere:

- muisinteractie voor leerlingen uitgeschakeld
- storende of niet-noodzakelijke knoppen verborgen of uitgeschakeld
- teksten aangepast aan een gestandaardiseerde onderzoeksofname

☐ Onderzoeksmodus inschakelen

Demo: metacognitie-uitleg
Kies of de demo ook het metacognitie-gedeelte (reflectie/uitleg) bevat.

☐ Demo bevat metacognitie-uitleg

Fixatiescherm
Toon voor elke oefening een fixatiekruis gedurende 1000 ms voordat de som verschijnt.

☒ Fixatiescherm inschakelen

Herhalingsregels
Deze regels bepalen hoe oefenvragen worden samengesteld. Ze helpen om herhaling en ongewenste patronen in de opgaven te vermijden.

☒ Voorkom identieke opgaven binnen één oefenset

☒ Vermijd opgaven met twee gelijke getallen (tie problems) (vb: 4+4, 7x7)

☐ Voorkom dezelfde uitkomst binnen één oefenset

☐ Commutativiteit: Beschouw omgekeerde opgaven als dezelfde (vb: behandel 3x7 en 7x3 als dezelfde opgave)

☒ Balans in positie van het grootste getal

Getallen uitsluiten
Komma-gescheiden lijst, bv. 0, 1, 11

Verboden waarden voor A (Eerste operand van de bewerking):

0,1

Verboden waarden voor B (Tweede operand van de bewerking):

0,1

Verboden antwoorden:

bijv. 0, 1, 10, 11

FIGURE 5.1: Research settings panel. Researchers can enable research mode, configure item generation rules (e.g. tie-problem exclusion, commutativity), define operand exclusion lists, and enable a fixation screen.

exercises but with different feedback conditions, or the same feedback conditions but with different difficulty settings, without writing any code.

5.2.3 Reproducibility

Section 2.2 defined reproducibility as the ability to save, share, and rerun the exact configuration of an experiment. RekenRangers supports this through a JSON-based export and import system, an idea suggested by Tijs Bellefroid. Every setting described in the previous section, from the operation selection to the research-mode rules to the exclusion lists, can be exported to a single JSON file with one click. A second researcher who wants to replicate the study can import that file into their own group, and every setting is restored exactly as it was. Sharing an experimental setup is then as simple as sharing a file.

One important consideration: two children configured with the same settings will not necessarily see the same sequence of exercises. The settings constrain which exercises are allowed, but the choice of which specific exercise comes next is partly random within those constraints. This is reproducibility at the level of experimental conditions, not at the level of literal trial-by-trial identity, which is the level that matters for between-group comparisons.

5.2.4 Flexibility

Flexibility, as defined in Section 2.2, is about whether a tool can support a wide variety of experimental designs without requiring the researcher to modify the underlying code. The settings system described in the previous sections already contributes to this: the ability to configure operations, difficulty modes, metacognitive prompts, timer settings, and item generation rules means that a single installation of RekenRangers can host studies with very different designs. But settings alone are not enough. Most research designs in educational psychology do not consist of a single uniform block of exercises. They involve phases: a warm-up, a main task, a break, a post-test, sometimes with different conditions across phases.

RekenRangers addresses this through an exercise sequence system. A sequence consists of multiple blocks, each with its own settings for the number of exercises, the operation, the difficulty mode, and whether metacognitive prompts are enabled. A researcher can, for example, configure a sequence that starts with 10 easy addition exercises as a warm-up without self-assessment, followed by three blocks of 15 adaptive exercises with metacognitive prompts, and a final block of 10 fixed-difficulty exercises as a post-test. Each block is independently configurable, and blocks can be added, edited, reordered, or removed through the dashboard. The research settings (research mode, exclusion rules, fixation screen) apply to the entire sequence rather than per block, which ensures consistency across phases.

The teacher dashboard also provides tools for managing progress through a sequence. The progress of individual students or the entire group can be reset to the beginning of the first block, which is useful when a session needs to be restarted or when the same sequence is used across multiple days. On the student side, a visual progress indicator shows the child where they are in the sequence, and how many blocks remain before they can puzzle. Figure 5.2 shows the sequence configuration screen in the teacher dashboard.

The sequence system tries to make RekenRangers more flexible than any of the exercise tools discussed in Chapter 3, and for the kinds of studies it was designed to support, the configurability is sufficient. It does not, however, match the flexibility of a general-purpose programming environment: studies that require trial structures outside the block model, custom adaptive algorithms, or stimulus types beyond arithmetic cannot be expressed within the current tool. This reflects what Lidwell, Holden, and Butler call the “flexibility–usability tradeoff”: “as the flexibility of a system increases, the usability of the system decreases” [28]. The limits it imposes on RekenRangers are discussed further in Section 8.2.

5.2.5 Reliability

Section 2.2 described reliability as having two dimensions: the tool must be robust, stable, and precise in its measurements, and it must keep running under real-world conditions on the kind of hardware schools actually have. RekenRangers addresses both.

On the robustness side, the platform is hosted on a KU Leuven server running

5.2. Approach: design decisions per property

#	Aantal vragen	Naam	Moedigheid	Operatie	Acties
1	10	Uit	Vast (1)	+	<button>Toevoegen</button> <button>Wijzig</button> <button>Verwijderen</button>
2	15	Aan	Adaptief	mix	<button>Toevoegen</button> <button>Wijzig</button> <button>Verwijderen</button>
3	15	Aan	Adaptief	mix	<button>Toevoegen</button> <button>Wijzig</button> <button>Verwijderen</button>
4	15	Aan	Adaptief	mix	<button>Toevoegen</button> <button>Wijzig</button> <button>Verwijderen</button>
5	10	Uit	Vast (5)	+	<button>Toevoegen</button> <button>Wijzig</button> <button>Verwijderen</button>

Reset voortgang leerlingen (Groep D)

Verwijder volledige reeks

Nieuw blok toevoegen

☐ Tijdsinstellingen

☒ Looftijd mag zelf kiezen of de timer zichtbaar is

Standaard tijdsinstelling

Beveiligingen

☒ Cirkelen (+)

☒ Vermengvragen (+)

☒ Afrekenen (-)

☒ Hks (zelfkeuring)

☒ Moedigheid

Moedigheidsmodus

Vaste moedigheid

Vaste moedigheid (indien van toepassing)

☒ Makkelijk

Aantal oefeningen in dit blok

☒ Heterogeniteit

No elke vraag

OK toevoegen

[Terug naar groep](#)

FIGURE 5.2: The exercise sequence configuration screen. Each row represents a block with its own settings. Blocks can be added, edited, or removed, and student progress can be reset individually or for the entire group.

inside a containerised environment using Podman and Docker, which isolates the application from the underlying system and makes deployment reproducible and stable. The MySQL database ensures that all data is persistently stored the moment an exercise is completed. If a child's browser crashes mid-session, or if the school's internet drops briefly, the data from every exercise completed up to that point is safe and does not need to be re-collected. The child can log back in and continue from where they left off without losing progress.

On the measurement side, response times are handled in JavaScript on the child's local machine. The timer for an exercise starts when the browser has fully rendered the stimulus on screen, and stops when the child submits their answer. This means that response times reflect actual thinking time rather than being inflated by network latency or server processing delays. The same local timing approach is used for the self-assessment and feedback phases.

The platform was tested across a range of common browsers and some small pilot sessions during development. A more substantial test of how RekenRangers holds up under real classroom conditions is provided by the experiment described in Chapters 6 and 7, with its outcomes discussed in Chapter 8. It is worth noting up front, however, that RekenRangers has not been benchmarked against the kind of formal millisecond-level timing study that has been carried out for tools like PsychoPy [10].

5.2.6 Ease of setup

The design principle that no feature should require programming knowledge runs through the entire teacher and researcher experience. Every setting discussed in the previous sections, from operation selection to research mode configuration to

exercise sequence design, is controlled through dropdown menus, checkboxes, toggle switches, and input fields in a graphical dashboard. There are no configuration files to edit, no scripts to write, and no command-line tools to learn. A researcher who wants to set up a study with three blocks of adaptive multiplication exercises with metacognitive prompts and tie-problem exclusion can do so in a few minutes without ever leaving the browser.

The exercise sequence system in particular benefits from its visual presentation. Each block in a sequence is displayed as a row in a table showing its key settings at a glance, which makes it easy to verify that the experimental design matches the intended protocol. Blocks can be added, reordered, or removed with a single click, and changes are saved immediately. This kind of visual configuration is one of the clearer differences with the research tools discussed in Chapter 2, where similar functionality would typically require writing and maintaining a Python script.

The teacher and researcher dashboard was designed to be self-explanatory, but for users who need additional guidance a comprehensive manual has been developed that explains every feature and setting in detail. The manual was drafted with the help of generative AI to speed up the writing, and then checked manually to make sure every described feature behaves as written. It is available both within the platform itself⁴ and as a standalone document included in Appendix D. A video walkthrough is also available.⁵

There is also room for improvement on this front, for example a preview of what the child will see given the current configuration, or a visual timeline of the exercise sequence rather than a table. The current setup interface is functional but does not yet scale with the number of options it now supports, which is one of the limitations of the platform discussed further in Section 8.2.

5.2.7 Adaptivity

A core requirement identified in Section 3.1 is that exercises should match the child's current ability level. RekenRangers implements this through an adaptive difficulty system inspired by the Elo rating principle, informed by the work of Klinkenberg et al. on the adaptive engine behind RekenRangers [27]. The system described below was developed in collaboration with Eline T'joen and Kiara Willems.

The central idea is that each child maintains a numerical rating that reflects their estimated skill level. This rating starts at 450 on a scale from 0 to 1000 and is updated after every exercise using the standard Elo update rule:

$$\theta'_j = \theta_j + K(S_{ij} - E(S_{ij})) \quad (5.1)$$

where θ_j is the child's current rating, S_{ij} is the score obtained on the exercise, and $E(S_{ij})$ is the expected score. The parameter K controls how strongly the rating

⁴https://rekenrangers.cs.kuleuven.be/static/handleiding_leerkracht.html

⁵<https://youtu.be/jeK2u8FNoPU>

reacts to each answer and was set to $K = 20$ after calibration across two pilot studies in the classroom settings.⁶

The expected score is fixed at $E(S_{ij}) = 0.75$. This value was chosen based on findings in the adaptive learning literature suggesting that a target success rate around 75% keeps learners in a productive zone: high enough to maintain confidence, but low enough that the task remains challenging [27]. The system converges toward a state where the child answers roughly three out of four exercises correctly.

How the actual score S_{ij} is computed depends on whether the child answered correctly. An incorrect answer always results in $S_{ij} = 0$, which produces a rating decrease. A correct answer incorporates response time through a quadratic scoring function:

$$S_{ij} = -\frac{5}{4d_i^2} \cdot t_{ij}^2 + 2 \quad (5.2)$$

where t_{ij} is the time the child took and d_i is the time limit for that exercise (15, 20, or 25 seconds for easy, medium, and difficult exercises respectively). This function has a natural interpretation. If the child answers exactly at the time limit, the score equals 0.75, which matches the expected score and produces no rating change. Answering faster yields a score above 0.75 and pushes the rating up. This means the system does not just track whether a child can solve a problem, but also how fluently they can solve it.

The rating then determines which exercises the child sees. Exercises are categorised into three difficulty levels (easy, medium, and difficult), and the system selects from a probability distribution that shifts with the rating, as shown in Table 5.1. A child below a rating of 250 receives mostly easy exercises, between 250 and 800 a mix dominated by medium exercises, and above 800 mostly difficult ones. Since children start at a rating of 450, they begin in the middle zone with a mix of all three levels. As they improve, the distribution shifts toward harder problems; if they struggle, it shifts back. The system is self-correcting and adjusts to each child individually, without the teacher needing to intervene.

What counts as easy, medium, or difficult depends on the operation and involves constraints on the operands (see Section 5.2.2), the result, and whether carrying or borrowing is required. The progression follows the same logic across operations: easy exercises stay within single-digit arithmetic, medium exercises introduce two-digit numbers without carrying or borrowing, and difficult exercises add that step. The full specification is provided in Appendix A.

It is worth noting that adaptivity in RekenRangers is configurable rather than fixed. In research mode, the teacher or researcher can disable the adaptive system and fix the difficulty at a specific level, which is essential for studies that require identical conditions across participants. This and the other research-control settings are discussed in Section 5.2.2.

⁶In the current version of RekenRangers, K is fixed at 20. Making this a configurable setting in the teacher dashboard is a natural next step.

Difficulty zone	Rating range	Exercise distribution
1	0 – 200	100% easy
2	200 – 400	70% easy, 30% medium
3	400 – 600	70% medium, 20% easy, 10% difficult
4	600 – 800	70% difficult, 30% medium
5	800 – 1000	100% difficult

TABLE 5.1: Difficulty zone distribution based on participant rating. Children are gradually exposed to harder exercises as their estimated ability increases.

5.2.8 Engagement

Section 3.1 identified engagement as the property that determines whether children actually want to use a tool, not just whether they can. For RekenRangers, this meant that simply presenting arithmetic exercises in a clean interface was not enough. The tool needed a reason for children to care, a narrative that gave the exercises meaning beyond getting the right answer.

The narrative developed for RekenRangers is built around a safari theme. The children are rangers (hence the name RekenRangers) and their mission is to rescue animals that have been captured by poachers. Solving arithmetic exercises earns coins, and coins are spent on puzzle pieces that gradually reveal an image of the animal. Once a puzzle is complete, the animal is freed. This framing turns every exercise session into a small step toward a visible goal, which gives even a routine practice session a sense of purpose.

The coin system is tied directly to performance. Children earn more coins for correct answers and for answering quickly, which creates a natural incentive to stay focused without the competitive pressure of leaderboards or peer comparison. The puzzles add a second layer of motivation: animals cost different amounts to rescue, so children have to choose which animal to work on next, and completing a more expensive puzzle feels like a bigger achievement. Figure 5.3 shows the puzzle selection screen.

The safari theme is not limited to the reward system. It runs through the entire application: the background illustrations, the button styles, the icons, and the login screen are all designed around the same visual identity. This consistency reinforces the idea that the child is inside a world, not just using a tool. All visual assets, including background scenes and animal images, were created using generative AI and then selected and adapted to fit the visual style of the application.

5.2.9 Feedback

After every exercise, the child receives immediate feedback on their answer. If the answer is correct, the screen shows a short confirmation message. If the answer is wrong, the correct answer is displayed alongside the child’s attempt, so that the child can see exactly where they went wrong. The feedback is deliberately kept

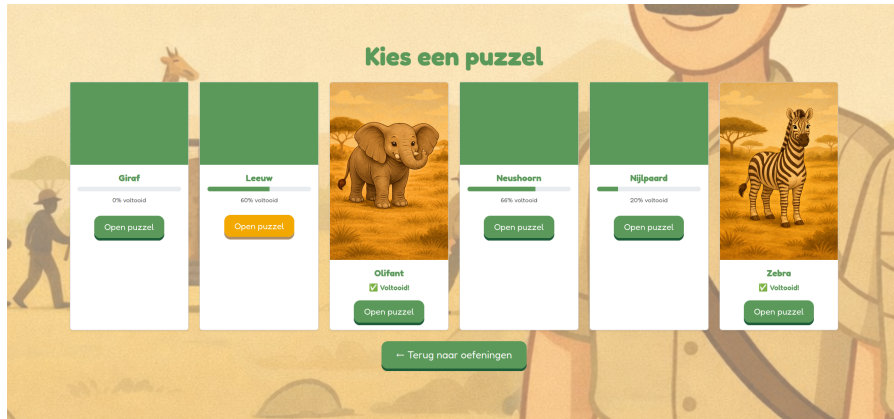


FIGURE 5.3: The RekenRangers puzzle environment.

simple and direct: no elaborate explanations or worked-out solutions, just a clear signal of correctness and, when needed, the right answer. This choice connects back to Section 3.1, where the literature on educational feedback suggested that minimal, focused feedback can support children’s persistence and strategy use at least as well as more elaborate variants. Extending this with worked-out explanations or hints could still be valuable, particularly for harder operations, and was kept simple here partly because the supported exercises (single-step addition, subtraction, and multiplication) did not require more and because there were time constraints on this project.

When the metacognitive self-assessment setting is enabled, a second layer of feedback is added. After each exercise, the child is asked to judge their own answer using a row of four smileys ranging from confident-correct (green) to confident-wrong (red). Once both the answer and the self-assessment are submitted, the feedback screen evaluates both. The self-assessment is considered correct if the child’s confidence matched the actual outcome: selecting green or yellow when the answer was right, or orange or red when it was wrong. If the child misjudged their own performance, the feedback tells them so explicitly. Figure 5.4 shows an example of an exercise that was answered incorrectly but where the child correctly anticipated the mistake.

The confidence judgement paradigm used here mirrors the one in the Bellon et al. study [6], and the smileys give it a visual language that is accessible to children as young as seven.

5.2.10 Personalisation

Personalisation is the property where RekenRangers has the most room for growth. The original intention was to integrate an avatar system into the safari theme, where each child would have their own ranger character that could be customised with outfits and accessories purchased with earned coins. Due to time constraints and the need to prioritise the research-critical features, this was not implemented

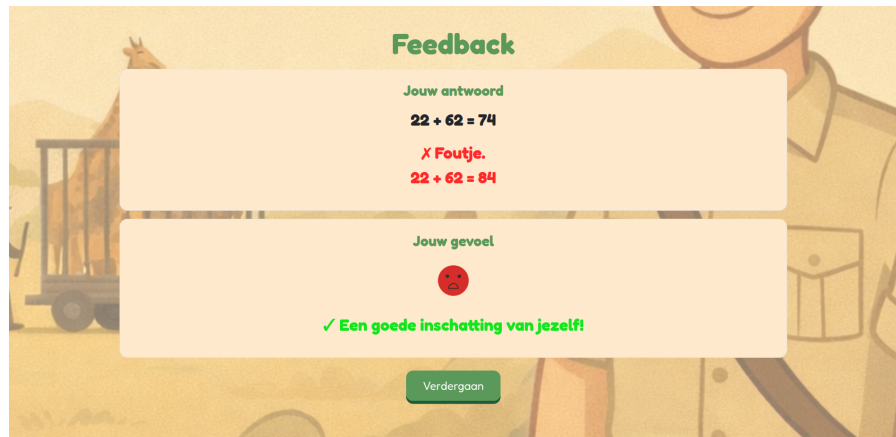


FIGURE 5.4: Feedback screen after an incorrectly answered exercise, where the child correctly anticipated the mistake through their self-assessment.

in the current version of the platform. The avatar gap is therefore one of the limitations discussed in Section 8.2, with concrete proposals listed under future work in Section 9.2.

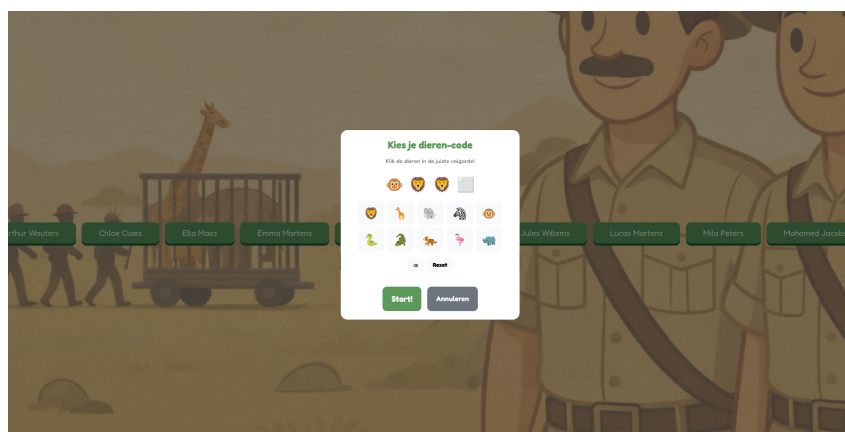
What is in place are the choices the platform gives the child. When enabled by the teacher or researcher, children can choose for themselves which operation they want to practise, what difficulty level they prefer, and whether they want to see a timer. As discussed in Section 3.1, this kind of choice can support a sense of ownership. Each child also works with their own account, so that their coin balance and puzzle progress are personal to them, and a child who has saved up coins and completed three animal puzzles sees a different puzzle screen than a classmate who just started. This gives some sense of individual identity within the platform, even without an avatar.

5.2.11 Accessibility

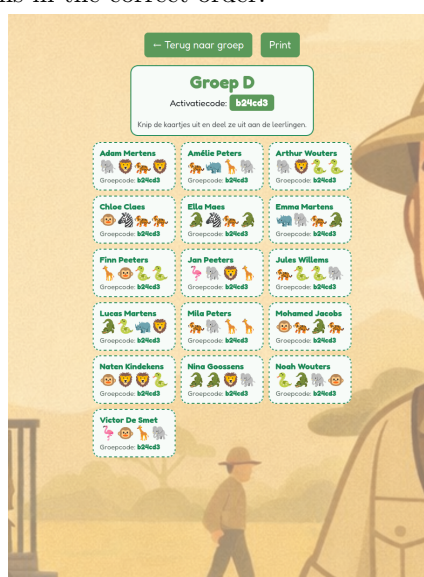
RekenRangers is a web application hosted on a KU Leuven server and accessible at <https://rekenrangers.cs.kuleuven.be/>. It runs in any modern browser without requiring software installation, plugins, or account creation through an external service. This means the tool works on the laptops, tablets, Chromebooks, and even phones that schools and families already have. The only requirements are an internet connection and a browser. There is currently no dedicated mobile app, and the interface is not specifically optimised for smaller screens, but both are realistic extensions for future development.

Access to the platform is managed through a two-step login system designed to be usable by young children without adult help. The child first enters a group code, which identifies their class or research group. They then select their name from a list and enter a personal animal code: a sequence of four animal icons that serves as a child-friendly alternative to a traditional password.

5.2. Approach: design decisions per property



(A) Student login screen. After selecting their name, the child enters a personal animal code by clicking four animal icons in the correct order.



(B) Printable login cards. Each card shows the child's name, group code, and personal animal code.

FIGURE 5.5: The RekenRangers login system.

Teachers and researchers generate these cards through the group management screen in the teacher dashboard. The cards can be printed and handed out in class, and since they contain everything a child needs to log in, they also enable use at home. Figure 5.5 shows the login screen and the printable cards together. This low-threshold approach to access means that RekenRangers can serve as both a classroom tool and a home practice tool without any additional setup.

5.2.12 User friendliness

The final exercise-tool property is user friendliness, and for RekenRangers this is less a single feature than a principle that runs through every screen of the application. The target audience is children between seven and eleven years old, and every interface element was designed with that in mind. Buttons are large and clearly labelled, navigation follows a simple top-to-bottom flow, and the visual language uses colours, icons, and illustrations rather than relying on text. All text that does appear is written in simple, child-appropriate Dutch.

To help children get started without needing an adult to explain the tool, RekenRangers includes an interactive demo mode. The demo walks the child through the full exercise flow step by step: how to read and answer an exercise, how to submit their answer, how the coin reward works, and how to navigate to the puzzle screen. When the metacognitive self-assessment setting is enabled, a separate version of the demo is available that also explains the smiley-based confidence judgement and what the feedback on it means. The demo can be replayed at any time, so a child who is unsure about something mid-session can revisit the explanation without having to ask for help.

One specific design decision worth mentioning is that the entire child-facing interface can be operated using only the keyboard. Answers are typed in directly, smileys are selected with the f, g, h, j keys, and puzzle pieces are bought with the spacebar. This choice was made in consultation with Elien Bellon, and is functionally closer to a piece of experimental control than to a usability feature: it ensures that when research mode is active and the mouse is disabled, every interaction the child needs is still reachable.

Across all of these choices, the goal is the one identified in Section 3.1: an interface that stays out of the child's way, so that the cognitive effort the child spends is spent on the arithmetic.

Chapter 6

Experimental Setup

The previous chapter described what RekenRangers is and how it addresses the twelve properties identified in Chapters 2 and 3. The question that remains is whether the tool actually delivers on these promises when used in a real classroom with real children. This chapter describes how that question was tested.

RekenRangers was deployed as the central training tool in a study conducted by Eline T’joen and Kiara Willems as part of their joint master’s thesis in pedagogical sciences at KU Leuven. The study involved 146 third-grade children (birth year 2017) across seven primary schools in Flanders, working with the platform over four training days in four different experimental conditions. Due to time limitations on the analysis side, the dataset used in this thesis was restricted to 107 of these participants. This deployment serves as the primary evaluation of RekenRangers: it tests whether the tool can support a real multi-session experiment while producing complete and reliable data (RQ2), and the questionnaire administered after the final session captures how the children experienced the platform while also providing useful feedback on the tool itself (RQ3). The details of the classroom deployment are described in Section 6.1, and the design of the questionnaire in Section 6.2.

6.1 Classroom deployments

The study, supervised by Dr. Elie Bellon and co-supervised by Prof. Dr. Bert De Smedt, investigated the effect of adaptive arithmetic training with metacognitive monitoring on the arithmetic skills and metacognitive abilities of third-grade children (ages 8–9). The participating schools were located in the provinces of Limburg and West-Flanders, and the data was collected during the spring of 2026.

Each condition involved four training days, with three exercise blocks of 18 exercises per day, totalling 54 exercises per session and 216 exercises across the full training period. All conditions used a mix of addition and multiplication, with an equal distribution of both operations by the end of each training day. The timer was hidden in all conditions. In the non-adaptive conditions (A and B), exercises were drawn in a semi-random order from a balanced mix of all three difficulty levels, with the constraint that no more than two consecutive exercises could come from the

same difficulty level. In the adaptive conditions (C and D), the Elo-based system described in Section 5.2.7 determined the difficulty distribution.

The research settings in RekenRangers were configured identically across all conditions: a fixation screen was shown before each exercise, identical exercises were blocked within a session, tie problems were excluded, commutativity was enforced, the position of the larger operand was balanced, and the values 0, 1, 10, and 11 were excluded as operands. These settings ensured that the generated exercises did not contain inherent biases caused by naturally easier solution strategies, allowing the evaluation to focus more fairly on the intended experimental conditions.

Prior to the main data collection, both researchers conducted a pilot study with one child each. The pilot confirmed that the adaptive algorithm produced appropriate difficulty progression, that the demo was understandable, and that the overall experience was positively received. One pilot participant reached the maximum Elo rating of 1000, indicating a ceiling effect, but after consultation this was attributed to the participant being an outlier rather than a structural problem with the difficulty range.

In addition to the RekenRangers training, participants completed a battery of pre- and post-tests, administered through OpenSesame and on paper, covering arithmetic, spelling, reasoning, and metacognitive monitoring. These measures were used by the other researchers to assess whether training effects transferred to standardised assessments and are not analysed in this thesis. After the final training session, children filled in the questionnaire described in Section 6.2.

6.2 The questionnaire

After the final session, each participating child filled in a short paper-based questionnaire about their experience with RekenRangers. The goal was to collect structured feedback on four aspects of the tool, each corresponding to a property defined earlier in this thesis: whether children found it motivating (engagement), whether they could use it without difficulty (user friendliness), whether it kept them engaged (engagement), and whether the feedback and difficulty felt appropriate (feedback and adaptivity).

Designing a questionnaire for children aged eight to nine years old imposes specific constraints. Questions must be short, concrete, and free of abstract formulations, and the response scale must be simple enough that a child can use it without hesitation. Research on surveying children confirms that these are methodological necessities, not just practical preferences: younger respondents are more sensitive to question complexity, and poorly formulated items produce unreliable data [9].

The questionnaire uses a 4-point Likert scale, where one star means “strongly disagree” and four stars mean “strongly agree”. A 4-point scale was chosen to discourage neutral midpoint responses and encourage clearer preference indications. The stars are presented visually on the paper form, making the scale intuitive without requiring explanation beyond a brief legend at the top of the page.

Three of the seven items are deliberately phrased in the negative direction. This is a standard technique in questionnaire design to detect “acquiescence bias”, the tendency of respondents to agree with every statement regardless of its content [41]. If a child gives four stars to both “I enjoyed using RekenRangers” and “I would not want to use RekenRangers again”, this signals inattentive or biased responding and allows the researcher to flag that response for closer inspection.

The items were not taken from an existing validated instrument, but are informed by established research traditions. The user friendliness items in particular reflect the core concerns of the System Usability Scale [13], simplified substantially to be appropriate for young children. The full questionnaire is included in Appendix E.

Chapter 7

Results

This chapter presents the results of the classroom deployment described in Chapter 6. It first establishes that the collected data is complete and reliable enough to draw conclusions from, then examines how the adaptive system behaved in practice, the engagement data from the coin and puzzle system, the children’s questionnaire responses, and finally the structured feedback from the researchers who used RekenRangers in the field.

7.1 Data quality and completeness

Across all four conditions, RekenRangers recorded 21,522 exercises from 102 participants. The original sample consisted of 107 participants, but five were excluded during data cleaning: four from condition A and one from condition B. The reason for exclusion was a bug discovered after the experiments, where participants who clicked the submit button very quickly could cause the same exercise to be recorded twice. Participants with a significant number of duplicate entries were removed; a few negligible duplicates in other participants were retained. The bug has since been fixed.

Not every participant completed all four training days; some children were absent on individual days due to illness or other reasons. Whether this matters depends on the analysis. Where each exercise is treated as an independent record, missing days do not affect the result. For analyses that depend on a participant’s full training history, such as the Elo trajectories, participants without all four days were excluded. On one occasion, the RekenRangers server was temporarily unavailable when a session was scheduled; the session was rescheduled and no data was lost.

The CSV export functioned as designed and could be loaded directly into data analysis frameworks without preprocessing, beyond the duplicate removal described above. One minor issue: server timestamps are offset due to a server-clock timezone setting, but this affects no analysis, since response times are measured and stored locally in the browser. Of the 102 participants, 51 completed the questionnaire described in Section 6.2; those results are presented in Section 7.4.

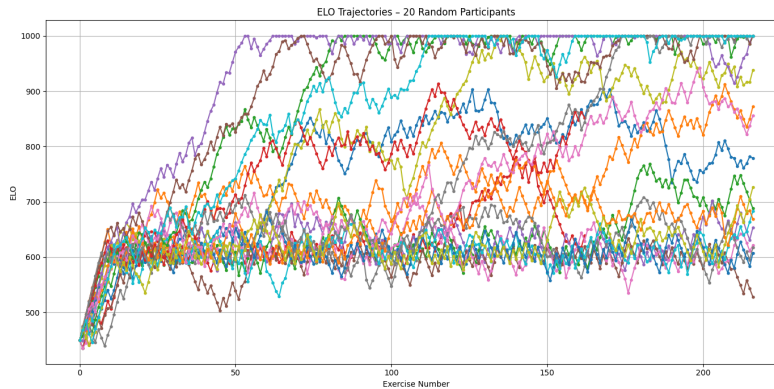


FIGURE 7.1: Elo rating trajectories of twenty randomly selected participants from the adaptive conditions. All participants started at a rating of 450. The trajectories illustrate how the system differentiates between children of different ability levels over the course of 216 exercises.

7.2 The adaptive system in practice

The adaptive difficulty system described in Section 5.2.7 was used in two of the four experimental conditions (C and D). The non-adaptive conditions (A and B) presented exercises drawn from a balanced mix of all three difficulty levels in a semi-random order. This section examines how the adaptive system behaved in practice using both quantitative data and qualitative observations.

7.2.1 Elo trajectories

Figure 7.1 shows the Elo trajectories of a larger sample of participants from the adaptive conditions over the course of their 216 exercises. All children started at the initial rating of 450, but their paths diverge quickly, and four broad patterns become visible across the sample.

First, the ceiling of 1000 was reached: a number of the strongest participants climbed rapidly during the first session and stayed at the ceiling for the rest of the training period. Second, and more striking, a sizeable subgroup of participants got stuck around a rating of 600. This value sits on the boundary between difficulty zones 3 and 4 (see Table 5.1), and the implications of this observation are discussed in detail in Chapter 8. Third, the trajectories are not smooth curves but jagged paths with frequent ups and downs, which is expected behaviour: the Elo system responds to every exercise, so a single mistake produces an immediate dip. The overall trend remains clearly readable in each case. Fourth, not every participant ended up at either the ceiling or the 600 plateau. A range of intermediate trajectories is also visible, which means the system is capturing real individual differences in ability rather than pushing everyone toward the same point.

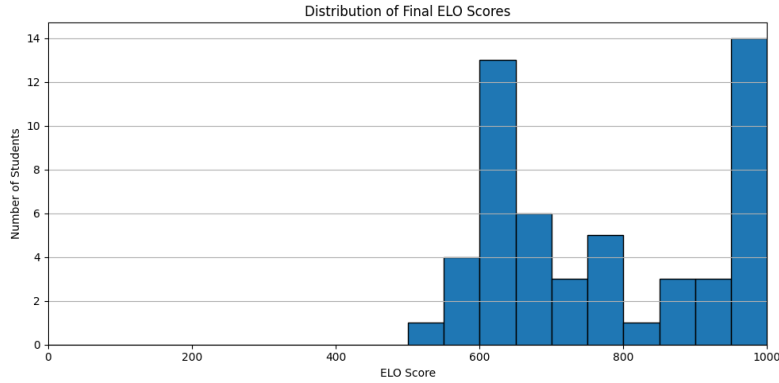


FIGURE 7.2: Distribution of final Elo scores across participants in the adaptive conditions.

7.2.2 Final Elo distribution

Figure 7.2 shows the distribution of final Elo scores across all participants in the adaptive conditions. The distribution spans from approximately 500 to 1000 and contains two distinct peaks. The first peak sits around 600–650 and is consistent with the pattern observed in the trajectories: a sizeable group of participants ended up oscillating around the boundary between difficulty zones 3 and 4, with their final rating reflecting the side of the boundary they happened to land on. The second peak is at the maximum rating of 1000 and represents children who reached the ceiling and stayed there. For these children, the exercise pool could no longer offer sufficient challenge, as all exercises were already drawn from the highest difficulty level.

7.2.3 Accuracy rates

Figure 7.3 compares the distribution of per-student accuracy rates between the adaptive and non-adaptive conditions. The non-adaptive group achieved a mean accuracy of 76% ($\sigma = 0.16$), which sits close to the 75% target. The adaptive group achieved a lower mean accuracy of 68% ($\sigma = 0.14$). The difference is expected: children in the adaptive conditions faced a higher average exercise difficulty (mean difficulty level of 2.54, compared to 2.00 in the non-adaptive conditions, where difficulty levels were balanced by design). The adaptive system pushed stronger children toward harder exercises, which increased the error rate. However, the adaptive system did not converge on the intended 75% target. The mean accuracy of 68% indicates that the current settings produced a difficulty level that was somewhat too challenging for the average participant. Why this happened, and what it would take to bring the system closer to its target, is examined in Section 8.1.

7. RESULTS

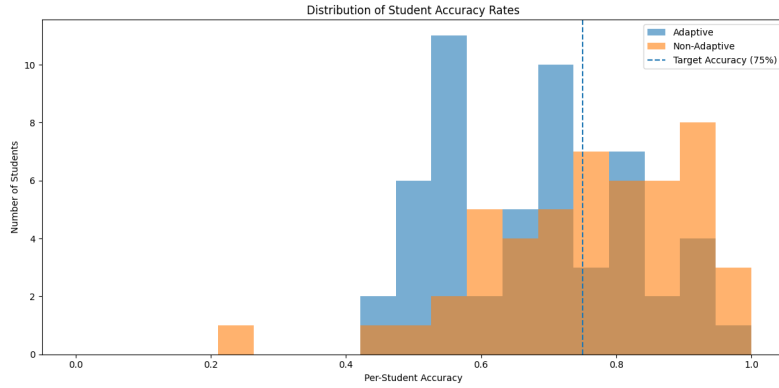


FIGURE 7.3: Distribution of per-student accuracy rates for the adaptive and non-adaptive conditions. The dashed line marks the 75% target accuracy. The adaptive group (mean 68%) falls below the target, while the non-adaptive group (mean 76%) lands close to it.

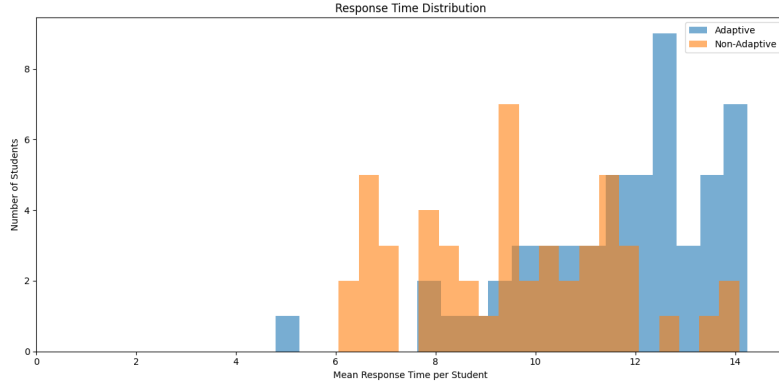


FIGURE 7.4: Distribution of mean response times per student for the adaptive (mean 11.7 s) and non-adaptive (mean 9.5 s) conditions. The adaptive group clusters toward longer response times, consistent with the scoring function’s equilibrium at the time limit.

7.2.4 Response times

Figure 7.4 shows the distribution of mean response times per student for both groups. The non-adaptive group has a mean response time of 9.5 seconds ($\sigma = 2.11$) and shows a wide, relatively uniform spread. This is consistent with the fact that these children received exercises from all three difficulty levels regardless of their ability: strong children solved easy exercises quickly while weaker children spent longer on the harder ones, producing a flatter distribution. The adaptive group has a higher mean response time of 11.7 seconds ($\sigma = 1.96$) and a narrower spread concentrated toward the longer end. This is consistent with how the scoring function works: its equilibrium point sits at the time limit, so the adaptive algorithm tends to push children toward exercises that take them somewhat longer to solve.

7.2.5 Qualitative observations

The quantitative patterns are accompanied by observations made during the classroom sessions. In their feedback report, the researchers noted that the non-adaptive group showed visible signs of frustration or boredom more frequently than the adaptive group. They also noted, particularly at one of the schools, that several children in the adaptive conditions asked why their exercises were so difficult, and that these children were all pupils who had not repeated a grade.

7.3 Engagement

7.3.1 Coins and puzzles

A central question of this thesis is whether gamification elements help keep children engaged across multiple sessions. The coin and puzzle system in RekenRangers offers a way to examine this with behavioural data rather than self-report: if children spend their coins on puzzles, they are engaging with the gamification system, and if they complete puzzles, they are treating the animal rescue mission as a genuine subgoal.

Table 7.1 summarises the coin and puzzle data across the four conditions. The results are striking: across the 102 participants, children spent on average 98.5% of their earned coins on puzzle pieces, and none spent less than 85%. The pattern holds across all four conditions, regardless of whether the condition included metacognitive prompts or adaptive difficulty. Children were never instructed to spend their coins: the puzzle shop was available at the end of each training day and using it was entirely optional, with no check or reward attached. The small number of unspent coins per child (on average 0 to 25 coins) is mostly explained by leftover coins being insufficient to buy another puzzle piece.

The puzzle completion data tells a similar story. Across all conditions, 99% of children completed at least one puzzle, and 44% completed three or more over the four training days. The coin economy was deliberately calibrated so that completing all six puzzles within 216 exercises was impossible, ensuring there was always a next goal to work toward. The data also shows different purchasing strategies: some children prioritised cheaper puzzles to rescue as many animals as possible, while others saved for more expensive puzzles and completed fewer.

The mean number of coins earned was higher in the non-adaptive conditions (A and B, around 665 coins) than in the adaptive conditions (C and D, around 615 coins). This follows from the adaptive system presenting harder exercises to stronger children, which lengthens response times and reduces the speed bonus. The spending percentages, however, were highly similar across all four conditions.

7.3.2 Within-day consistency

A second way to look at engagement is to ask how stable performance stays within a training day. Section 2.5 showed that in a comparable OpenSesame training, children answered progressively faster while becoming less accurate as the day went

7. RESULTS

Condition	n	Mean earned	Mean spent	Mean % spent	Mean puzzles completed
A	23	646.0	634.1	98.2	2.3
B	26	683.2	671.3	98.2	2.7
C	27	610.4	601.9	98.6	2.0
D	26	620.7	613.5	98.8	2.0

TABLE 7.1: Average coin earning, spending, and puzzle completion across the four experimental conditions.

on, a pattern consistent with disengagement. The same view can be applied to the RekenRangers data. Each training day consisted of three exercise blocks of 18 exercises (Section 6.1), giving twelve blocks across the four days, labelled S1 to S12 in the figures.

Figure 7.5 shows the average correctness and response time per block for the non-adaptive conditions (A and B), the closest match to the no-adaptivity setup in Section 2.5. The equivalent figures for the adaptive conditions (C and D) are given in Appendix F.

What stands out is mainly what does not happen. Correctness stays in a fairly narrow band across all twelve blocks, without the steady within-day decline that marked the OpenSesame data in Section 2.5. Response times drop over the first three blocks and then settle, staying roughly flat for the rest of the training. Eyeballing the patterns, the RekenRangers blocks look more consistent than Baeten’s, though this is only a visual impression: the samples and designs differ, and no formal comparison is made here. The interpretation of these patterns is taken up in Chapter 8.

7.4 Questionnaire results

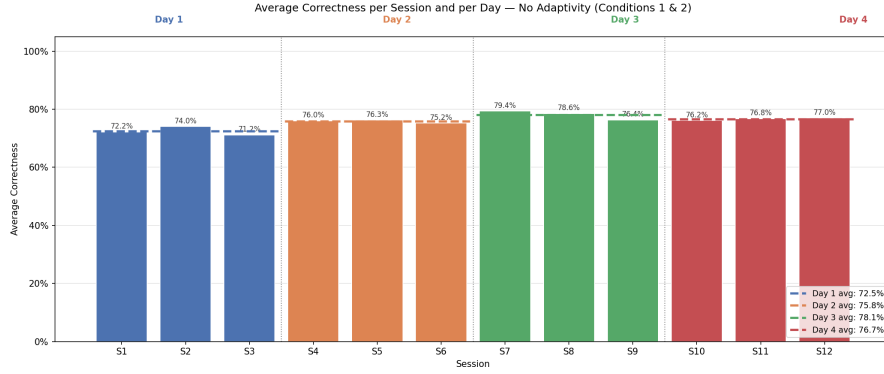
Of the 51 questionnaires returned, 45 were retained for analysis. Six questionnaires were excluded because they were either filled in incompletely or because all questions received an identical response.¹ Table 7.2 summarises the responses across the 45 valid questionnaires. The full text of the questions is available in Appendix E. The interpretation of these results is presented in Section 8.1.

7.5 Researcher feedback

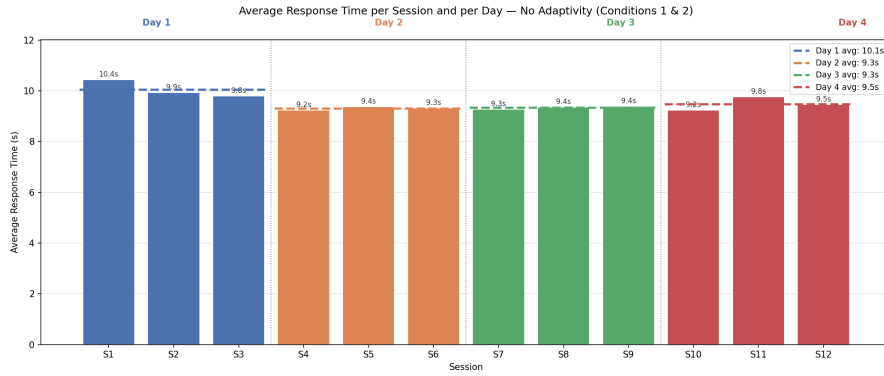
After the classroom deployments, Eline T’joen and Kiara Willems provided structured written feedback on their experience using RekenRangers as a research tool.²

¹The incomplete questionnaires were likely caused by the recto-verso printing format, where children only filled in the front side of the page.

²Both researchers used RekenRangers for their joint master’s thesis in pedagogical sciences, and submitted their feedback together in a single document. Because both were involved in the development process and provided input on the tool’s design throughout, their perspective is that of informed insiders rather than first-time users. This should be taken into account when interpreting the feedback.



(A) Average correctness per block.



(B) Average response time per block.

FIGURE 7.5: Average correctness and response time per block for the non-adaptive conditions (A and B), across the twelve blocks of the four training days.

Question	1 star	2 stars	3 stars	4 stars	Weighted mean
Q1 (positive)	0	0	10	35	3.78
Q2 (negative)	28	9	5	3	1.62
Q3 (positive)	0	1	8	36	3.78
Q4 (negative)	21	11	12	1	1.84
Q5 (positive)	1	4	18	22	3.36
Q6 (positive)	0	7	17	21	3.31
Q7 (negative)	8	13	18	6	2.49

TABLE 7.2: Questionnaire results based on 45 valid responses. Items marked as negative are phrased in the negative direction, so lower scores indicate a more positive evaluation of the tool. Responses are on a 4-point scale where 1 means "strongly disagree" and 4 means "strongly agree." The full text of each question is provided in Appendix E.

The feedback is organised around four aspects: setting up an experiment, running it, monitoring it, and the quality of the collected data, followed by a set of additional suggestions.

On **setting up an experiment**, the researchers reported that configuring the study in RekenRangers was straightforward once the available options were understood. The initial learning curve was comparable to any new tool, and the manual helped navigate the interface. Compared to their experience with PsychoPy, RekenRangers made the setup process less complex: after consulting the literature to decide on the desired settings, the configuration could be done by selecting options rather than writing code. They also noted that the tool gives a clear warning when combinations of settings are incompatible and would generate insufficient exercises, though they suggested that the message could be more specific about which setting is causing the conflict.

On **running the experiment**, the researchers described the experience as a clear improvement over OpenSesame. They reported that OpenSesame regularly produced errors when opening tasks, particularly after software updates, and that these errors were time-consuming to resolve. By contrast, RekenRangers was easy to start and rarely showed errors. When an issue did occur, it was quickly resolved and the tool continued without further problems. They did note a minor bug where the settings had to be re-saved before each training day, and that the interactive demo was too long for the children, particularly the version that included the metacognitive explanation, causing visible drops in enthusiasm.

On **monitoring the experiment**, the researchers found it useful that they could check individual participant data directly within the tool through the results view, without needing to download the full dataset first. They suggested that the set identifiers in the export could be numbered per participant (e.g., 1 through 12) rather than using global database identifiers, which would make it easier to see at a glance which training day a particular record belongs to. They also suggested adding a live progress view so that a teacher or researcher could see in real time how far each child is in their exercise sequence.

On **data quality**, the researchers assessed the collected data as good, with the main concern being the duplicate entries caused by the submission bug described in Section 7.1. They highlighted the convenience of being able to download the data directly from the researcher's account in the browser, without needing to transfer files between devices, and noted that knowing the data was stored server-side and could be recovered in case of problems was reassuring.

Finally, the researchers offered several **additional suggestions**: displaying the logged-in child's name on screen after login to help verify that each child is working on the correct account, showing the number of participants per group on the dashboard to help balance conditions, and giving the RekenRangers logo a place on the front page.

Chapter 8

Discussion

This chapter interprets what the results of the classroom deployment mean for the central question of this thesis. It first works through the findings from Chapter 7, then sets out the limitations of both the tool and the evaluation, and closes with a personal reflection on building RekenRangers in collaboration with educational scientists.

8.1 Interpretation of results

The central question behind this thesis is whether a single tool can bridge the gap between classroom exercise tools and controlled research software. The results from the classroom deployment provide a first answer. This section interprets the findings from Chapter 7 along four dimensions: whether the tool worked as a research instrument, how the adaptive difficulty system behaved, whether the engagement system delivered, and how the end users experienced the platform.

Did the tool work as a research instrument? The short answer is yes. Two researchers with no programming background configured a 2×2 factorial study with four conditions, deployed it across seven schools, ran it over four training days with 107 participants, monitored progress through the dashboard, and exported structured data ready for analysis. There were problems along the way (the duplicate submission bug, a server downtime incident, a demo that was too long) but none of these resulted in permanent data loss or forced the study to be abandoned. The researchers' feedback confirms that the overall experience was a clear improvement over their previous work with OpenSesame and PsychoPy, particularly in terms of setup complexity and data accessibility. The final scientific results of the study are not yet available, as Eline T'joen and Kiara Willems will complete their master's thesis in the next academic year. But the fact that the tool carried a study of this scale to completion, producing 21,522 clean exercise records across four conditions, demonstrates that the gap identified in Chapters 2 and 3 can be addressed in practice.

Did the adaptivity system perform as designed? Partially. The system succeeded at its primary goal: differentiating between children of different ability levels and adjusting the difficulty accordingly. Final Elo scores spread from 500 to 1000, the trajectory data shows the system responding individually to each child, and children in the adaptive conditions received exercises that were on average harder (mean difficulty 2.54) than those in the non-adaptive conditions (2.00). Where the system fell short is in its calibration. The intended target accuracy of 75% was not reached; the adaptive group averaged 68%, meaning exercises were somewhat too difficult for the average participant. This quantitative finding is echoed in the qualitative observations: children in the adaptive conditions asked why their exercises were so hard, which is the kind of reaction the 75% target was meant to avoid. The most likely causes are the fixed expected score and the quadratic scoring function (Equation 5.2), both of which were calibrated on limited pilot data. A more refined approach would assign an Elo rating to each individual exercise rather than relying on predefined difficulty categories, following the item-level model used in Rekenluin [27].

A second calibration issue is visible in the distribution of final Elo scores. A clear peak appears around 600–650, which is the boundary between difficulty zones 3 and 4 (see Table 5.1). The trajectory plots confirm that many of these children were oscillating across this boundary rather than progressing through it. The most likely explanation is the size of the step between zone 3 and zone 4: in zone 3 only 10% of presented exercises are difficult, while in zone 4 this jumps to 70%. The moment a child's rating crosses 600, the proportion of difficult exercises increases sharply, the child gets more answers wrong, the scoring function pulls the rating back below 600, and the cycle repeats. A more gradual transition between zones, or a continuous mapping from rating to difficulty mix rather than a step function, would likely smooth this out.

A separate observation concerns response times. The adaptive group clustered toward longer response times while the non-adaptive group was spread uniformly, which is what the scoring function predicts when its equilibrium sits at the time limit. The pattern confirms that the system behaves as designed, but it raises the question of whether working at the edge of the time limit is pedagogically desirable. A scoring function with a more generous equilibrium might produce a more comfortable experience without sacrificing the adaptive mechanism.

A fourth observation concerns the rating ceiling. Fourteen children reached the maximum rating of 1000, meaning the exercise pool could no longer challenge them. This, the 600 plateau, and the 68% accuracy are all calibration and content problems, not architectural ones. The adaptive framework responds to performance, differentiates between children, and adjusts in real time. What it needs is better parameters, a more gradual zone transition, and a deeper exercise pool, all of which are discussed under future work in Section 9.2.

Did the engagement system deliver? The behavioural data suggests it did. Children spent 98.5% of their earned coins on puzzle pieces, and 99% completed at

least one puzzle, all without any instruction or incentive to do so¹. If the gamification had been irrelevant, most coins would have gone unspent; instead, the near-complete spending and the variation in purchasing strategies (cheap puzzles for quantity versus expensive ones for the challenge) point to genuine, voluntary engagement with the reward system. The researchers' qualitative observations support this: children were visibly enthusiastic about rescuing animals and asked to continue playing.

The important concern is that engagement is not the same as learning. Research on gamification in education has consistently found positive effects on motivation and engagement, but the evidence for direct improvements in learning outcomes remains mixed [43]. The data here shows that children were motivated, not that the motivation translated into better arithmetic. A first informal look at the training data is consistent with this caution: across the four days, average accuracy increased slightly and average response time decreased slightly, but no statistical testing was carried out, and to the eye both measures look fairly constant. Establishing a genuine causal link between gamification and learning outcomes would require a study designed specifically for that purpose, which was outside the scope of this thesis.

How did the end users experience RekenRangers? The questionnaire results paint a clear picture from the children's side. The three motivation items (Q1, Q2 and Q5) all received highly positive scores, with weighted means of 3.78, 1.62 (negatively phrased), and 3.36 respectively. Children clearly enjoyed working with RekenRangers and would want to use it again, which reinforces the behavioural engagement evidence from the coin and puzzle data. The user friendliness items (Q3 and Q4) scored similarly strongly: children understood what they had to do (mean 3.78) and rarely needed help to operate the tool (mean 1.84 on the negatively phrased item). The feedback item (Q6) also received a high score (3.31), suggesting that the simple correct/incorrect signal was clear enough for the children to interpret. The adaptivity item (Q7) is the most divided, with a mean of 2.49: a substantial portion of the children felt that the exercises did not match their level well. This aligns with the quantitative finding that the adaptive system did not converge on its 75% accuracy target and was somewhat too challenging on average.² Together, the questionnaire results confirm what the behavioural data already suggested: children experienced RekenRangers as engaging and easy to use, and the main area for improvement remains the calibration of the adaptive system.

¹Although the children were informed that puzzles were available and could be explored after each training session, they were never required, reminded, or externally rewarded to interact with them.

²Due to logistic constraints during the deployment, the returned questionnaires could not be traced back to individual experimental conditions. As a result, it is not possible to compare the responses on Q7 between the adaptive and non-adaptive groups, which would have provided a sharper test of whether the perceived mismatch comes specifically from the adaptive conditions.

8.2 Limitations

This section discusses the main limitations in two parts: limitations of RekenRangers as a tool, and limitations of the evaluation presented in this thesis. .

8.2.1 Limitations of the tool

The most significant limitation of RekenRangers is its restricted flexibility. As discussed in Section 5.2.4, there is an inherent tension between making a tool accessible to non-programmers (ease of setup) and supporting the full range of experimental designs a researcher might want to run (flexibility) [28]. RekenRangers sits firmly on the accessibility side of that trade-off. The study by Eline T’joen and Kiara Willems was possible because the platform was developed with their specific experimental design in mind, but for studies that require a custom adaptive algorithm, a different stimulus modality, or trial structures outside the block-based model, RekenRangers offers no path forward. The tool is, in its current form, a research instrument for arithmetic studies that fit its built-in templates, not a general experiment builder. Addressing this without sacrificing the no-code principle is a non-trivial design problem and is discussed further in Section 9.2.

A related limitation concerns the teacher and researcher interface. While the configuration is performed entirely through dropdown menus and checkboxes, the interface itself was not designed for the scale it now supports. As the number of options grew over the course of development, the structure of the settings screens did not scale with it, and the result is a panel that is functional but lacks the visual organisation a user might expect from a mature research tool. The researchers’ feedback specifically noted that they cannot see a preview of the experiment they are configuring. There is no equivalent of the structured experiment view that OpenSesame and PsychoPy offer. This is a real usability gap, and it limits the practical experience of setting up complex studies.

Several of the twelve properties identified in Chapters 2 and 3 are present but underdeveloped. Personalisation is the clearest example: the foundation for an avatar system, customisable profiles, and personal statistics is in place, but no visual personalisation was implemented within the scope of this thesis. The feedback system is deliberately minimal, providing only correctness information; as discussed in Section 3.1, minimal feedback can be a sound choice, but extending it with worked-out explanations or strategy hints could still be valuable. Adaptivity, finally, works as a mechanism but is not yet well calibrated, as the analysis in Section 8.1 showed. All three are realistic targets for future development and are discussed in Section 9.2.

Finally, there are smaller operational limitations worth mentioning. There is currently no way for a researcher or teacher who has forgotten their password to recover it without contacting the administrator. This illustrates the kind of practical detail that becomes important when a tool moves from prototype to deployment.

8.2.2 Limitations of the evaluation

The evaluation presented in this thesis has limitations of its own. The most significant is that the only researchers who used RekenRangers to run a real study were Eline T'joen and Kiara Willems, both of whom were closely involved in the design of the tool throughout its development. Their feedback is informative and detailed, but they are not first-time users. Some of the design choices in the platform were made specifically in response to their input, which means their experience is unlikely to be representative of what a researcher encountering the tool for the first time would find. A meaningful external evaluation, with researchers and teachers who had no prior contact with the project, would be necessary to confirm the conclusions drawn here.

The thesis also does not provide direct evidence about learning outcomes. The data presented in Chapter 7 shows that children engaged with the tool, that the adaptive system differentiated between ability levels, and that the platform collected complete and structured data. What it cannot show is whether children who used RekenRangers improved their arithmetic skills more than they would have using another tool. Answering that question would require a dedicated study designed for that purpose, which falls outside the scope of the present work.

8.3 Collaborating across disciplines: reflections for a computer scientist

Building a tool like RekenRangers in collaboration with educational scientists is a tougher challenge than I expected at the start of this thesis. Computer scientists are trained to think in terms of features, systems, and abstractions. Educational scientists are trained to think in terms of variables, conditions, and confounds. They see things that I would never have thought about, and they ask precise questions that often expose hidden assumptions in the design. Working with my co-supervisor Elien Bellon and with Eline T'joen and Kiara Willems was an enormous benefit to the quality of the tool: they grounded design decisions in actual research practice, they brought credibility to the project, and they pushed me to think more carefully than I would have on my own.

It also made the development process considerably more iterative. Almost every part of RekenRangers, from the front-end interface to the adaptive algorithm, went through several rounds of feedback and compromise. The collaboration was rewarding but not always efficient.

The tension worked in both directions. From my side, I sometimes underestimated how much detail educational research demands: a setting that seemed acceptable to me would turn out to have implications for confounding variables that I had not considered. From the other side, Elien, Eline, and Kiara naturally framed their requirements in terms of their own studies. My goal was to build a more general tool, which meant I often had to translate a specific feature request into a more abstract version that would also serve other researchers in the future. This was not

always easy, and it required active pushback from my side to keep the tool from becoming a custom-built instrument for one experiment.

Miscommunication occasionally led to wasted effort. The clearest example was a long struggle over the "randomness" of exercise selection. To me, random meant statistically random. To them, it meant varied in a way that avoided unwanted patterns. We spent weeks aligning on what the word actually meant before the implementation could be finalised. Small misalignments like this, when accumulated, slow the project down considerably.

If I had to point to one lesson, it would be the importance of explicitly discussing the design goals up front, and revisiting them throughout the project. It is tempting to dive into features and implementation details from the start, but a few hours spent agreeing on what the tool is for, and for whom, would have saved many more hours of rework. Cross-disciplinary collaboration is genuinely valuable, but it requires both sides to keep their long-term goals visible to each other.

Chapter 9

Conclusion and Future Work

This final chapter draws the thesis together. It first revisits the three research questions and the answers the work provides, then sets out the most promising directions for future development.

9.1 Conclusion

This thesis set out to address a clear gap in the digital tools available for arithmetic education in primary schools. Classroom exercise tools provide the engagement, adaptivity, and child-friendly design that keep children motivated, but lack the data tracking, experimental control and flexibility needed for scientific research. Research tools, on the other hand, provide the scientific infrastructure but lack the motivational qualities needed to keep children engaged across multiple classroom sessions. No existing tool covered both sides. RekenRangers was built to fill that gap.

Three research questions structured the work:

- RQ1** Can a single web-based platform satisfy both the exercise-tool properties and the research-tool properties identified in this thesis?
- RQ2** Can the platform support a real multi-session classroom experiment with over 100 participants across multiple conditions, producing complete and reliable data, keeping children engaged, and running an adaptive difficulty system that performs as designed?
- RQ3** How do children evaluate the engagement and user friendliness of the platform, and how do teachers and researchers evaluate the ease of setup?

Answer to RQ1 RekenRangers brings the twelve properties together in a single tool, but not all of them to the same extent. Figure 9.1 summarises how the platform delivers on each property. Data tracking, experimental control, reproducibility, engagement, accessibility, and user friendliness are fully realised: the tool collects

9. CONCLUSION AND FUTURE WORK

complete and structured data, gives researchers fine-grained control over every aspect of an experiment, supports reproducibility through JSON export and import, offers a coherent and motivating game environment, runs in any browser without installation, and presents an interface that children can operate independently. Personalisation, adaptivity, reliability, and feedback are present and functional but could be extended. Personalisation lacks an avatar system, the adaptive algorithm requires further calibration, the feedback system is minimal, and the reliability of the platform, while sufficient for the classroom deployment, was affected by bugs and a server downtime incident that revealed room for improvement. Flexibility remains the clearest limitation: the tool supports a wide range of arithmetic studies through its settings and sequence system, but cannot match the open-ended configurability of a programming language. Ease of setup is more difficult to assess. The configuration is performed entirely without code, which is a major step forward, but the interface itself needs further work and an evaluation with researchers who were not involved in the development. In short: RekenRangers shows that bringing both worlds together is possible, but the trade-off between flexibility and ease of setup remains a real one, and several of the twelve properties leave room for improvement.

	Data tracking	Experimental control	Reproducibility	Flexibility	Reliability	Ease of setup	Adaptivity	Engagement	Feedback	Personalisation	Accessibility	User friendliness
	✓	✓	✓	●	✓	?	✓	✓	✓	✓	✓	✓

FIGURE 9.1: Final evaluation of RekenRangers against the twelve properties from Chapters 2 and 3. Full check marks indicate properties that are fully realised, half check marks indicate properties that are functional but could be extended, an orange dot indicates a fundamental trade-off, and a question mark indicates a property that needs further evaluation.

Answer to RQ2 For the most part, yes. RekenRangers supported a multi-session classroom experiment with 107 participants across four conditions and four training days, deployed in seven primary schools. The platform collected 21,522 exercise records, the data was structured and exportable without preprocessing, and no data was lost despite a brief server downtime incident during the deployment. The engagement data shows that children spent 98.5% of their earned coins voluntarily and that 99% of participants completed at least one puzzle. The adaptive system differentiated between ability levels, but it did not reach its target accuracy, hit the rating ceiling for 14 children, and left a sizeable subgroup oscillating around the zone 3–4 boundary. These results demonstrate that RekenRangers can carry a real study from setup to data export and keep children engaged throughout, but they also point to clear calibration work that the adaptive system still needs.

Answer to RQ3 The two researchers who used the tool, Eline T’joen and Kiara Willems, evaluated the ease of setup positively, particularly in comparison to their

previous experience with OpenSesame and PsychoPy. The setup was manageable, the data was easy to retrieve, and the tool ran reliably during the classroom sessions. The most important concern, as discussed in Section 8.2, is that both researchers were closely involved in the development of the tool, which limits the generalisability of their feedback.

From the children’s side, the questionnaire results indicate that both engagement and user friendliness were evaluated positively, while the perceived match between the exercises and the children’s ability level remains an area for improvement.

Closing remarks RekenRangers does not aim to replace either category of existing tools entirely. Its purpose is to offer an alternative for arithmetic studies that need to take place in real classroom settings, where engagement, ecological validity, and data collection have to coexist. The work presented here shows that this combination is achievable, while making the remaining trade-offs and open problems explicit. Whether RekenRangers becomes a genuine instrument for the educational research community depends on how those open problems are addressed. The next section outlines the most important directions for future work.

9.2 Future work

Several directions for future work follow directly from the limitations identified in Section 8.2.

The most important next step is to evaluate RekenRangers in terms of actual learning outcomes. The current thesis demonstrated that the tool collects clean data and keeps children engaged, but it could not show whether using RekenRangers produces measurable improvements in arithmetic skills or metacognitive awareness. A dedicated learning study, ideally with a control condition using a more traditional tool, would close this gap. The data from the deployment by Eline T’joen and Kiara Willems will provide a first answer once their thesis is completed.

Several of the twelve properties should be extended. Personalisation deserves a visible avatar system and a personal statistics dashboard, both of which the existing account infrastructure already supports. The feedback system could move beyond a binary correct/incorrect signal toward short explanations or strategy hints. The adaptive algorithm needs further parameter tuning, a more gradual transition between difficulty zones, and a scoring function with a more generous equilibrium. It would also benefit from an item-level Elo model in the style of RekenRangers [27], together with a deeper exercise pool to remove the ceiling effect observed in the deployment.

The flexibility of the tool remains its most significant constraint. Future work should explore ways to expand the range of supported experimental designs without reintroducing the programming barrier. Possible directions include a domain-specific configuration language, a modular block system that allows researchers to define new exercise types declaratively, or the use of generative AI to translate a researcher’s

natural-language description of a study into the appropriate configuration. Each of these comes with its own trade-offs and is worth investigating.

A meaningful external evaluation is also needed. The current feedback comes exclusively from researchers who were involved in the development. Testing the tool with researchers and teachers who have no prior connection to the project would give a much clearer picture of its real-world usability and would likely reveal ease of setup issues that the current evaluation could not detect.

Finally, if RekenRangers is to grow beyond a research prototype, the underlying infrastructure needs to be brought to a more professional level. This includes a cleaner and better-documented codebase, a more robust hosting setup, automated testing, proper user account recovery flows, and the kind of monitoring and maintenance procedures that any production-grade educational platform requires. These are not glamorous engineering tasks, but they are what separates a working prototype from a tool that researchers and schools can rely on.

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Appendix A

Exercise Specification

RekenRangers supports three arithmetic operations: addition, subtraction, and multiplication. For each operation, exercises are classified into three difficulty levels based on properties of the operands and the result. This appendix describes the classification rules for each operation.

A.1 Addition

The difficulty of an addition exercise $a + b = s$ is determined by the magnitude of the operands and whether the addition requires carrying (i.e., whether the sum of the unit digits exceeds or equals 10).

Level	Criteria
1 (easy)	The sum is less than 10. Both operands are single-digit numbers and the result stays within the single digits.
2 (medium)	The sum is 10 or greater, but no carrying is required, except for two single digits. This includes cases where both operands are single-digit and the sum just crosses 10, as well as cases involving two-digit operands where the unit digits sum to less than 10.
3 (difficult)	Both operands are two-digit numbers and carrying is required: the sum of the unit digits is 10 or greater.

TABLE A.1: Difficulty classification for addition exercises.

A.2 Subtraction

The difficulty of a subtraction exercise $a - b$ (where $a \geq b$) depends on the magnitude of the operands and whether borrowing is required (i.e., whether the unit digit of a is smaller than the unit digit of b).

A. EXERCISE SPECIFICATION

Level	Criteria
1 (easy)	Both operands are single-digit numbers and the result is less than 10.
2 (medium)	At least one operand is a two-digit number, but no borrowing is required: the unit digit of a is greater than or equal to the unit digit of b .
3 (difficult)	Both operands are two-digit numbers and borrowing is required: the unit digit of a is smaller than the unit digit of b .

TABLE A.2: Difficulty classification for subtraction exercises.

A.3 Multiplication

The difficulty of a multiplication exercise $a \times b = p$ is based on the magnitude of the operands and the product.

Level	Criteria
1 (easy)	The product is at most 25. These are small multiplication facts that children typically memorise early.
2 (medium)	Both operands are at most 10 (single-digit multiplication tables), but the product exceeds 25.
3 (difficult)	Both operands are at most 20, with exactly one operand exceeding 10. These exercises require multiplying a single-digit number by a two-digit number.

TABLE A.3: Difficulty classification for multiplication exercises.

These classification rules are implemented as deterministic functions in the back-end. When the adaptive system selects an exercise, it first determines which difficulty level to draw from (based on the child's current rating and the distribution in Table 5.1), then generates a random exercise that satisfies the corresponding criteria.

Appendix B

Data Fields

Table B.1 lists all fields recorded for each exercise completed in RekenRangers.

Field	Description
user	The student who completed the exercise
set_id	Unique identifier linking all exercises completed in one session
operation	The arithmetic operation (addition, subtraction, or multiplication)
left_operand	The first operand of the exercise
right_operand	The second operand of the exercise
user_answer	The answer entered by the child
real_answer	The correct answer
is_correct	Whether the child's answer was correct
response_time	Time taken to answer the exercise (in milliseconds)
timestamp	The exact date and time the exercise was completed
difficulty	The difficulty level of the exercise (1, 2, or 3)
timer_visible	Whether the timer was visible to the child during the exercise
confidence_judgement	The self-assessment smiley selected by the child (if enabled)
confidence_calibration	Whether the self-assessment matched the actual outcome
response_time_selfscore	Time taken to select the confidence judgement
response_time_feedback	Time spent on the feedback screen
current_elo	The child's Elo rating at the time of the exercise
delta_elo	The change in Elo rating produced by this exercise
difficulty_zone	The difficulty zone the child was in (1 through 5)

TABLE B.1: Complete list of data fields recorded for each exercise in RekenRangers.

Appendix C

Group Settings

Table C.1 lists all configurable settings available in the RekenRangers teacher dashboard for each group.

C. GROUP SETTINGS

Setting	Description
<i>General settings</i>	
Timer mode	Whether children can choose to show or hide the timer.
Timer visibility	Whether the timer is forced on or off.
Available operations	Which operations are available: addition, subtraction, multiplication, or a mix
Mix composition	When mix is selected, which operations are included
Difficulty mode	Adaptive (Elo-based), fixed by the researcher, or chosen by the child
Fixed difficulty level	When fixed, the difficulty level (1, 2, 3, or a mix)
Metacognitive prompting	Whether the self-assessment smiley prompt is shown after each exercise
<i>Research settings</i>	
Research mode	Disables mouse interaction, hides non-essential buttons, and standardises interface text
Demo with metacognition	Whether the interactive demo includes the self-assessment explanation
Fixation screen	Whether a fixation cross is displayed before each exercise
Prevent identical exercises	Blocks the same exercise from appearing twice in one session
Exclude tie problems	Prevents exercises where both operands are equal
Prevent repeated results	Blocks exercises with the same outcome from appearing consecutively
Commutativity	Treats commuted exercises (e.g., 3×7 and 7×3) as identical
Operand position balance	Balances whether the larger operand appears on the left or right
Excluded values (operand A)	List of values that may not appear as the first operand
Excluded values (operand B)	List of values that may not appear as the second operand
Excluded values (result)	List of values that may not appear as the answer

TABLE C.1: Complete list of group settings available in the RekenRangers teacher dashboard.

Appendix D

Teacher and researcher manual



Handleiding voor Leerkrachten & Onderzoekers

Versie 1.0 · Mei 2026

Inhoudstafel

1. Inleiding
2. Account aanmaken & inloggen
 - 2.1 Registreren als leerkracht
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3. Leerkracht Dashboard
 - 3.1 Overzicht van het dashboard
 - 3.2 Nieuwe groep aanmaken
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4. Groepspagina
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6. Oefenreeks beheren
 - 6.1 Wat is een oefenreeks?
 - 6.2 Blok toevoegen
 - 6.3 Blok aanpassen of verwijderen
 - 6.4 Voortgang leerlingen resetten
 - 6.5 Volledige reeks verwijderen
7. Resultaten van een leerling bekijken
8. Naam van een leerling wijzigen
9. CSV-exportbestand – kolomuitleg
10. Verklarende woordenlijst

1. Inleiding

RekenRangers is een webapplicatie waarmee leerlingen rekenopgaven (optellen, vermenigvuldigen, aftrekken) inoefenen. De applicatie is speciaal ontworpen voor gebruik in onderwijskundig onderzoek en biedt leerkrachten en onderzoekers uitgebreide mogelijkheden om de afname te controleren en de resultaten te raadplegen.

Dit document beschrijft *alle* functies die beschikbaar zijn via het leerkrachtportaal. Leerlingen loggen in met een groepscode en pincode; de leerkracht beheert via het portaal de groepen, instellingen, oefenreeksen en resultaten.

Leerlingenlogin

Leerlingen loggen **niet** in via de leerkrachtpagina. Zij gebruiken de knop "*Inloggen als leerling*" op de startpagina en voeren de groepscode + pincode in die op hun leerlingenkaartje staat.

2. Account aanmaken & inloggen

2.1 Registreren als leerkracht

Bezoek de startpagina en klik op "**Inloggen als leerkracht**", gevolgd door "**Nog geen account? Registreer!**". Vul het formulier in:

Veld	Beschrijving
Gebruikersnaam	Unieke naam voor uw account (maximaal 255 tekens).
Wachtwoord	Kies een sterk wachtwoord. Klik op het oog-icoon om het zichtbaar te maken tijdens het typen.

Klik op "**Account Aanmaken**". Als de gebruikersnaam al bestaat, verschijnt een foutmelding en kunt u een andere naam kiezen.

Wachtwoord vergeten?

Er is geen automatische wachtwoordherstelflow. Neem contact op met de systeembeheerder als u geen toegang meer heeft.

2.2 Inloggen als leerkracht

Ga naar de startpagina → "**Inloggen als leerkracht**". Vul uw gebruikersnaam en wachtwoord in en klik op "**Login**". Na een succesvolle login wordt u doorgestuurd naar het *Leerkracht Dashboard*.

3. Leerkracht Dashboard

3.1 Overzicht van het dashboard

Het dashboard is het centrale vertrekpunt. Bovenaan vindt u het formulier om een nieuwe groep aan te maken en een sectie om leerlingen via CSV in te laden. Daaronder ziet u een tabel met alle bestaande groepen.

3.2 Nieuwe groep aanmaken

- 1 Vul een groepsnaam in het tekstveld in (bijv. "Klas 4A").
- 2 Klik op "**Aanmaken**".
- 3 De groep verschijnt direct in de tabel met een automatisch gegenereerde **activatiecode** (6 tekens, hoofdletters/cijfers).

Activatiecode

Elke groep heeft een unieke activatiecode. Leerlingen voeren deze code in bij het inloggen om aan de juiste groep te worden gekoppeld. U kunt de code altijd terugvinden op de groepspagina.

Elke leerkracht heeft zijn/haar eigen groepen. Groepsnamen moeten uniek zijn per leerkracht (twee leerkrachten mogen wel dezelfde naam gebruiken).

3.3 Leerlingen importeren via CSV

U kunt meerdere leerlingen tegelijk importeren via een CSV-bestand. Dit is handig bij de start van een schooljaar.

Vereist bestandsformaat

Het bestand moet een CSV zijn met de volgende kolommen (met koprij):

Kolom	Inhoud
Voornaam	Voornaam van de leerling
Achternaam	Achternaam van de leerling
Groepsnaam	Naam van de groep waartoe de leerling behoort

Elke unieke groepsnaam in de CSV wordt automatisch aangemaakt als die nog niet bestaat. De volledige naam van de leerling wordt samengesteld als *Voornaam + spatie + Achternaam*. Elke leerling krijgt automatisch een willekeurige pincode van 4 cijfers.

- 1 Klik op "**Bestand kiezen**" en selecteer uw CSV-bestand.
- 2 Klik op "**Upload en importeer**".
- 3 Leerlingen die al bestaan in de groep worden automatisch overgeslagen.

Let op de encoding

Sla het CSV-bestand op in **UTF-8**-codering om problemen met speciale tekens (é, è, ij...) te vermijden.

3.4 Bestaande groepen beheren

De groepentabel toont per groep: naam, activatiecode, aanmaakdatum en twee acties:

- **Open** – ga naar de gedetailleerde groepspagina.
- **Verwijder** – verwijdert de groep *inclusief alle leerlingen, oefeningen en resultaten*. Er verschijnt een bevestigingsdialoog. Deze actie is **onomkeerbaar**.

4. Groepspagina

De groepspagina is het hart van het beheer. U bereikt haar via *Dashboard* → *Open* naast een groep. De pagina toont:

- Groepsnaam & activatiecode.
- Formulier om een nieuwe leerling toe te voegen.
- Tabel met alle leerlingen in de groep.
- Actieknoppen voor export, afdrukken en oefenreeksbeheer.
- Formulier voor groepsinstellingen (zie hoofdstuk 5).

4.1 Leerling manueel toevoegen

- 1 Typ de naam van de leerling in het tekstveld.
- 2 Klik op "**Toevoegen**".

De leerling krijgt automatisch een willekeurige pincode van 4 cijfers. Als de naam al bestaat in deze groep, verschijnt een foutmelding.

4.2 Leerlingenlijst lezen

Kolom	Betekenis
Naam	Naam van de leerling.
Pincode	De 4-cijferige pincode die de leerling nodig heeft bij het inloggen.
Reeksvoortgang	Voortgangsbalk die toont hoeveel blokken van de oefenreeks de leerling heeft voltooid. – Geel = bezig; Groen = klaar; "Geen reeks actief" als er geen reeks is aangemaakt. – De tekst toont <i>voltooid / totaal</i> (bijv. 2 / 5).
ELO-rating	De huidige ELO-score van de leerling (start op 450). Wordt gebruikt bij adaptieve moeilijkheid. Een hogere waarde betekent betere prestaties.
Bekijk resultaten	Knop om de gedetailleerde oefenhistoriek van de leerling te openen.
Reset reeksvoortgang	Zet de voortgang van <i>deze leerling</i> terug naar stap 1 (bevestiging vereist).
Verwijderen	Verwijdert de leerling en <i>alle</i> bijhorende resultaten permanent (bevestiging vereist).

4.3 Leerlingenkaartjes afdrukken

Klik op "**Print leerlingenkaartjes**". Er opent een nieuwe pagina met een kaartenraster (3 per rij). Elk kaartje toont:

- Naam van de leerling.
- Een reeks emoji-dieren als geheugensteuntje voor de pincode (de pincode zelf is ook leesbaar op het

kaartje).

- De groepscode.

Klik op **"Print"** om te afdrukken of als PDF op te slaan. De kaartjes zijn ontworpen om uitgeknipt te worden.

💡 **Emoji-pincodes**

Elke pincode wordt omgezet naar een reeks dier-emoji's. Dit maakt het voor jonge leerlingen gemakkelijker om hun code te onthouden. Voorbeeld: 4721 → 🦊🐘🐧🦁

4.4 Exporteer groepsdata naar CSV

Klik op **"Exporteer groep naar CSV"**. Het bestand wordt onmiddellijk gedownload en bevat alle oefenresultaten van alle leerlingen in de groep. Zie hoofdstuk 9 voor een gedetailleerde uitleg van alle kolommen.

5. Groepsinstellingen

Onderaan de groepspagina bevindt zich het instellingenformulier. Wanneer er een **actieve oefenreeks** is (zie hoofdstuk 6), zijn de gewone groepsinstellingen verborgen en kunt u enkel nog de *Onderzoeksinstellingen* aanpassen. De oefenreeks neemt dan de controle over bewerkingen, moeilijkheid, etc.

5.1 Timerinstellingen

Instelling	Beschrijving
Leerling kiest timer	Vinkje aan → de leerling mag zelf kiezen of de reactietijdtimer zichtbaar is. Vinkje uit → de instelling hieronder wordt dwingend opgelegd.
Standaard timerinstelling	<i>Zichtbaar</i> – de timer wordt getoond. <i>Verborgen</i> – de timer wordt niet getoond (de tijd wordt nog steeds geregistreerd in de achtergrond).

5.2 Bewerkingen

U kiest welke bewerkingstypen de leerling mag uitvoeren:

Optie	Beschrijving
Optellen (+)	Somopgaven, bijv. $6 + 9$.
Vermenigvuldigen (×)	Maaltafelopgaven, bijv. 7×8 .
Aftrekken (–)	Aftrekopgaven, bijv. $15 - 7$.
Mix (willekeurig)	Als dit aangevinkt is, kan de leerling kiezen voor een willekeurige mix van bewerkingen. Een extra sectie " <i>Samenstelling van Mix</i> " verschijnt: hier geeft u aan welke bewerkingen in de mix kunnen voorkomen. Minstens één optie moet aangevinkt zijn.

Minstens één bewerking verplicht

Als u alle bewerkingen uitvinkt, kan de instelling niet worden opgeslagen.

5.3 Moeilijkheid

Modus	Beschrijving
Vaste moeilijkheid	Alle opgaven hebben de moeilijkheid die u instelt in het veld " <i>Vaste moeilijkheid</i> ". Keuzes: ★ Makkelijk, ★★ Gemiddeld, ★★★ Moeilijk, of Mix (willekeurig per opgave).
Leerling kiest zelf	De leerling selecteert vóór elke reeks zelf de gewenste moeilijkheidsgraad.

Adaptief (ELO)	De moeilijkheid wordt automatisch aangepast op basis van de ELO-score van de leerling. Goed presterende leerlingen krijgen moeilijkere opgaven; leerlingen die het moeilijk hebben, krijgen makkelijker opgaven.
-----------------------	--

5.4 Aantal oefeningen per reeks

Stel in hoeveel opgaven één oefensessie bevat. Minimaal 1, maximaal 50. Standaard: 10. Veelgebruikte waarden: 5, 10, 15.

5.5 Metacognitie

Optie	Beschrijving
Uit	Geen reflectiemoment na de opgave.
Na elke vraag	Na elke opgave krijgt de leerling een zelfbeoordelvraag: hoe zeker was je van je antwoord? (schaal 1–4). Deze score wordt geregistreerd als <code>meta_self_score</code> .

5.6 Onderzoeksinstellingen

Deze sectie is altijd zichtbaar, ook wanneer er een oefenreeks actief is.

Onderzoeksmodus

Schakel **Onderzoeksmodus inschakelen** in voor een gestandaardiseerde onderzoeksafname:

- Muisinteractie voor leerlingen is uitgeschakeld.
- Niet-noodzakelijke knoppen worden verborgen of uitgeschakeld.
- Teksten worden aangepast voor een neutrale onderzoekscontext.

Demo: metacognitie-uitleg

Bij het eerste inloggen krijgt elke leerling een **demo** te zien die uitlegt hoe de applicatie werkt. Met "**Demo bevat metacognitie-uitleg**" kiest u of ook het metacognitiedeel (de zelfbeoordeling) in de demo uitgelegd wordt. Zet dit aan als u metacognitie inschakelt, zodat leerlingen weten wat ze moeten doen.

Fixatiescherm

Als dit aangevinkt is, verschijnt er vóór elke opgave gedurende 1000 ms een fixatiekruis (+) in het midden van het scherm. Dit dient om de aandacht van de leerling te focussen vóór de som verschijnt, een standaardprocedure in cognitief-psychologisch onderzoek.

Herhalingsregels

Regel	Beschrijving
Voorkom identieke opgaven binnen één oefenset	Dezelfde combinatie van bewerking + a + b mag niet twee keer voorkomen in dezelfde sessie.
Vermijd tie problems	Sluit opgaven uit waarbij beide operanden gelijk zijn (bijv. $4+4$, 7×7).
Voorkom dezelfde uitkomst binnen één oefenset	Twee opgaven met hetzelfde antwoord mogen niet in dezelfde sessie voorkomen.
Commutativiteit: omgekeerde opgaven als hetzelfde beschouwen	3×7 en 7×3 worden als één en dezelfde opgave beschouwd. Relevant in combinatie met "geen duplicaten".
Balans in positie van het grootste getal	Zorgt ervoor dat het grootste getal ongeveer even vaak links als rechts staat (bijv. $7+3$ en $3+7$ worden afgewisseld).

Getallen uitsluiten

U kunt bepaalde waarden verbieden voor de operanden of het antwoord. Geef een kommagescheiden lijst in:

Veld	Beschrijving
Verboden waarden voor A	Eerste operand mag deze waarden niet aannemen. Bijv. 0, 1, 11
Verboden waarden voor B	Tweede operand mag deze waarden niet aannemen.

Gebruik voor replicateerbaar onderzoek

Door dezelfde combinatie van uitsluitingsregels en herhalingsregels te gebruiken in meerdere groepen, zorgt u voor vergelijkbare en reproduceerbare opgavensets.

5.7 Instellingen opslaan, exporteren & importeren

Opslaan

- 1 Klik op " **Instellingen opslaan**" rechtsonder in het formulier.
- 2 De applicatie valideert de instellingen en bouwt de oefenpool voor. Dit kan tot een minuut duren bij strenge herhalingsregels; een laadscherm wordt getoond.

Lange verwerkingstijd mogelijk

Het opslaan van instellingen triggert het voorbereiden van alle mogelijke oefencombinaties. Sluit de pagina niet tijdens het laadscherm.

Voorbeeldopgave bekijken

Klik op "**Bekijk voorbeeldopgave**" om te controleren of de instellingen de gewenste opgaven genereren, vóór leerlingen beginnen.

Instellingen exporteren

Klik op " **Exporteer instellingen**". Er wordt een JSON-bestand gedownload met alle groepsinstellingen én de oefenreeks (indien aanwezig). Dit bestand kunt u opslaan als sjabloon of delen met collega's.

Instellingen importeren

- 1 Klik op "**Bestand kiezen**" naast het importveld en selecteer een eerder geëxporteerd JSON-bestand.
- 2 Klik op " **Importeer instellingen**" en bevestig de melding.

Alle huidige instellingen van de groep worden overschreven. Als het bestand een oefenreeks bevat, wordt deze ook geïmporteerd en worden alle leerlingen teruggezet naar stap 1.

Let op

Importeren overschrijft de volledige configuratie van de groep. Dit kan niet ongedaan worden gemaakt. Exporteer de huidige instellingen vooraf als back-up indien nodig.

6. Oefenreeks beheren

6.1 Wat is een oefenreeks?

Een **oefenreeks** bestaat uit meerdere *blokken* die leerlingen **na elkaar** doorlopen. Elk blok heeft zijn eigen instellingen voor bewerkingen, moeilijkheid, timer, metacognitie en aantal oefeningen. Zo kunt u een gestructureerde leerlijn opbouwen, bijv.:

- Blok 1: optellen, makkelijk, geen metacognitie, 10 opgaven.
- Blok 2: optellen, gemiddeld, metacognitie aan, 10 opgaven.
- Blok 3: mix van optellen en vermenigvuldigen, adaptief, metacognitie aan, 15 opgaven.

De **onderzoeksinstellingen** (uitsluitingsregels, fixatiekruis, ...) gelden altijd voor de gehele groep en worden niet per blok ingesteld.

Wanneer er een actieve oefenreeks is, vervangen de blokinstellingen de gewone groepsinstellingen voor bewerkingen, moeilijkheid, timer, metacognitie en aantal opgaven.

Bereik de reekspagina via de groepspagina → " **Beheer oefenreeks voor deze groep**".

6.2 Blok toevoegen

Onderaan de reekspagina staat het formulier "*Nieuw blok toevoegen*". De instellingen zijn identiek aan de gewone groepsinstellingen (timer, bewerkingen, moeilijkheid, aantal oefeningen, metacognitie), maar nu specifiek voor dit blok:

Sectie	Opties
Timerinstellingen	Zie §5.1
Bewerkingen	Zie §5.2 — minstens één bewerking verplicht
Moeilijkheid	Zie §5.3
Aantal oefeningen	1–50; standaard 10
Metacognitie	Uit of na elke vraag

Klik op " **Blok toevoegen**". Het blok verschijnt in de tabel bovenaan met zijn positienummer.

6.3 Overzichtstabel van blokken

De tabel bovenaan de reekspagina toont per blok:

Kolom	Beschrijving
#	Positienummer (volgorde van uitvoering).
Aantal Vragen	Opgaven per sessie in dit blok.
Meta	Aan of Uit voor metacognitivraag.

Moeilijkheid	<i>Adaptief</i> = ELO-gebaseerd <i>Leerling kiest</i> = vrije keuze <i>Vast (n)</i> = vaste ster-waarde (1, 2, 3 of Mix)
Operaties	Symbolen voor de ingeschakelde bewerkingen: +, ×, −, mix.
Acties	Knoppen voor aanpassen en verwijderen.

6.4 Blok aanpassen

Klik op **"Aanpassen"** naast een blok. U komt op een bewerkingspagina met dezelfde velden als bij het aanmaken. Pas de gewenste waarden aan en klik op  **Opslaan**. Klik op **"← Terug"** om te annuleren.

6.5 Blok verwijderen

Klik op **"Verwijderen"** naast een blok en bevestig de melding. De overige blokken worden automatisch hernummerd (1, 2, 3, ...). Als het laatste blok wordt verwijderd, valt de groep terug op de standaard groepsinstellingen.

6.6 Voortgang leerlingen resetten (stap 1)

Klik op **"Reset voortgang leerlingen (stap 1)"** en bevestig. *Alle* leerlingen in de groep beginnen opnieuw vanaf blok 1. Hun resultaten worden **niet** gewist.



Wanneer gebruiken?

Gebruik deze knop aan het begin van een nieuwe onderzoeksmeting waarbij alle leerlingen opnieuw de volledige reeks moeten doorlopen.

6.7 Volledige reeks verwijderen

Klik op **"Verwijder volledige reeks"** en bevestig. Alle blokken worden verwijderd en leerlingen worden automatisch teruggezet naar stap 1. De groep gebruikt vanaf dan weer de standaard groepsinstellingen. Resultaten van afgewerkte blokken blijven behouden.

7. Resultaten van een leerling bekijken

Klik op " **Bekijk**" naast een leerling in de groepstabel. U ziet een tabel met alle afzonderlijke oefeningen die de leerling ooit heeft gemaakt, gesorteerd op reeks en volgorde.

Kolom	Beschrijving
Reeks	Het set-nummer (elke voltooide oefensessie is één set).
Operatie	Symbool van de bewerking: + (optellen), × (vermenigvuldigen), – (aftrekken).
Moeilijkheid	★ = makkelijk, ★★ = gemiddeld, ★★★ = moeilijk. Een streepje (–) betekent geen niveau bekend.
Timer	Ja = timer was zichtbaar; Nee = timer was verborgen.
a	Eerste operand van de bewerking.
b	Tweede operand van de bewerking.
Antwoord	Het antwoord dat de leerling heeft ingevoerd.
Correct	Het correcte antwoord.
Juist?	✓ (groen) = correct antwoord; ✗ (rood) = fout antwoord.
Reactietijd (s)	De tijd in seconden tussen het verschijnen van de opgave en het indienen van het antwoord.
Meta_score	De zelfbeoordelingsscore (1–4) die de leerling heeft gegeven na de opgave, als metacognitie aanstond. 0 = geen metacognitiemoment (of niet beantwoord).

Boven de tabel staat de naam van de leerling en de groep waartoe hij/zij behoort.

8. Naam van een leerling wijzigen

Op de resultatenpagina van een leerling staat een knop " **Naam wijzigen**". Klik erop om een invoerveld te tonen:

- 1 Klik op " **Naam wijzigen**".
- 2 Pas de naam aan in het tekstveld.
- 3 Klik op "**Opslaan**" of "**Annuleren**".

Uniekheidsregel

De nieuwe naam mag niet al bestaan bij een andere leerling in dezelfde groep.

9. CSV-exportbestand – kolomuitleg

De export via "**Exporteer groep naar CSV**" bevat één rij per afzonderlijke oefening, voor alle leerlingen in de groep. Het bestand is UTF-8 gecodeerd met BOM (voor compatibiliteit met Excel).

#	Kolomnaam	Beschrijving
1	<code>username</code>	Naam van de leerling.
2	<code>set_id</code>	Numerieke identificatie van de oefensessie. Alle rijen met hetzelfde <code>set_id</code> horen bij dezelfde sessie.
3	<code>operation</code>	Bewerking: <code>add</code> , <code>multiply</code> of <code>negation</code> .
4	<code>a</code>	Eerste operand.
5	<code>b</code>	Tweede operand.
6	<code>user_answer</code>	Het door de leerling ingevoerde antwoord.
7	<code>correct_answer</code>	Het correcte antwoord op de opgave.
8	<code>is_correct</code>	1 = correct, 0 = fout.
9	<code>response_time_exercises</code>	Reactietijd in seconden voor het beantwoorden van de som.
10	<code>exercise_timestamp</code>	Datum en tijdstip waarop de oefening werd gemaakt (ISO-formaat).
11	<code>exercise_difficulty</code>	Moeilijkheidsgraad: 1 = makkelijk, 2 = gemiddeld, 3 = moeilijk.
12	<code>show_timer</code>	1 = timer was zichtbaar, 0 = verborgen.
13	<code>confidence</code>	Zelfbeoordelingsscore van de leerling (1–4). 0 of leeg = geen metacognitie.
14	<code>calibration_of_confidence</code>	Kalibratie-index: mate waarin het zelfvertrouwen overeenkomt met de prestatie. 3 = perfect gekalibreerd (zeker + correct, of onzeker + fout). 2 = goede kalibratie. 1 = redelijke kalibratie. 0 = slechte kalibratie (zeker + fout, of onzeker + correct). <i>Leeg als er geen metacognitiemoment was.</i>
15	<code>response_time_self_score</code>	Reactietijd in seconden voor het invullen van de zelfbeoordeling (metacognitie).
16	<code>response_time_feedback</code>	Reactietijd in seconden voor het bekijken van de feedbackpagina na de opgave.
17	<code>current_elo</code>	ELO-score van de leerling op het moment van de opgave.

18	<code>delta_elo</code>	Verandering in ELO-score na deze opgave (positief = stijging, negatief = daling).
19	<code>current_difficulty_zone</code>	De moeilijkheidszone (1–3) die het adaptieve systeem op dat moment toeweest.

💡 Kalibratie-index berekening

De kalibratie-index (`calibration_of_confidence`) wordt als volgt bepaald:

- Score 4 (zeker) + correct antwoord → kalibratie = 3
- Score 3 (tamelijk zeker) + correct → kalibratie = 2
- Score 2 (tamelijk onzeker) + correct → kalibratie = 1
- Score 1 (onzeker) + correct → kalibratie = 0
- Omgekeerd voor foute antwoorden (score 1 + fout = 3, score 4 + fout = 0).

10. Verklarende woordenlijst

Term	Betekenis
Activatiecode	Unieke 6-tekens code per groep. Leerlingen voeren deze in om in te loggen in de juiste groep.
Adaptieve moeilijkheid (ELO)	Een systeem waarbij de moeilijkheidsgraad automatisch wordt aangepast op basis van de prestaties van de leerling, vergelijkbaar met het ELO-systeem in schaken.
Blok	Eén stap binnen een oefenreeks met eigen instellingen.
Commutativiteit	De eigenschap dat $a+b = b+a$ en $a \times b = b \times a$. Als "beschouw omgekeerde opgaven als dezelfde" aanstaat, worden 3×7 en 7×3 als duplicaten beschouwd.
ELO-score	Een getal dat de vaardigheid van een leerling weergeeft. Begint op 450. Stijgt bij correcte antwoorden, daalt bij foute. Wordt gebruikt bij adaptieve moeilijkheid.
Fixatiescherm	Een fixatiekruis dat 1 seconde voor elke opgave verschijnt om de aandacht te focussen.
Groepscode	Synoniem voor activatiecode.
Metacognitie	Reflectie over het eigen leerproces. In deze app: een zelfbeoordelvraag na elke opgave ("Hoe zeker was je?").
Mix (bewerking)	Een optie waarbij per opgave willekeurig een bewerking wordt gekozen uit de door de leerkracht geselecteerde mix-bewerkingen.
Oefenpool	De voorberekende lijst van geldige opgaven voor een groep, op basis van de ingestelde regels. Wordt herbouwd bij elke instellingswijziging.
Oefenreeks	Een geordende reeks blokken die leerlingen stap voor stap doorlopen.
Operand	Een getal in een rekensom. In " $6 + 9$ ": 6 is operand A, 9 is operand B.
Onderzoeksmodus	Een speciaal afnameprofiel dat de interface neutraliseert voor gecontroleerd wetenschappelijk onderzoek.
Pincode	Een 4-cijferige code die elke leerling krijgt. Leerlingen voeren deze in samen met hun naam en de groepscode om in te loggen.
Set	Eén voltooide oefensessie. Alle opgaven binnen een set hebben hetzelfde <code>set_id</code> in de database.
Tie problem	Een opgave waarbij beide operanden gelijk zijn: $4+4$, 7×7 , etc.

Bijlage A – Workflow samenvatting

Eerste keer instellen (eenvoudige afname)

1. Registreer een leerkrachtaccount.
2. Maak een groep aan op het dashboard.
3. Voeg leerlingen toe (manueel of via CSV-import).
4. Stel de groepsinstellingen in (bewerkingen, moeilijkheid, timer, metacognitie).
5. Stel de onderzoeksinstellingen in (herhalingsregels, exclusielijsten, ...).
6. Klik "**Instellingen opslaan**" en wacht op het laadscherm.
7. Druk leerlingenkaartjes af en deel ze uit.
8. Leerlingen loggen in via de groepscode op hun kaartje.

Eerste keer instellen (oefenreeks)

1. Voer stappen 1–5 hierboven uit.
2. Ga naar " **Beheer oefenreeks**".
3. Voeg alle gewenste blokken toe met hun specifieke instellingen.
4. Ga terug naar de groepspagina en sla de onderzoeksinstellingen op.
5. Deel leerlingenkaartjes uit.

Tijdens de afname

- Volg de reeksvoortgang via de tabel op de groepspagina.
- Bekijk tussentijdse resultaten per leerling via " **Bekijk**".
- Reset indien nodig individuele leerlingen of de hele groep via de resetknoppen.

Na de afname

- Download de volledige dataset via "**Exporteer groep naar CSV**".
- Exporteer de instellingen als back-up of sjabloon via " **Exporteer instellingen**".



Appendix E

Questionnaire

RekenRangers

Vragenlijst

Voor jou

We willen graag weten wat jij van RekenRangers vond.

Lees elke zin goed. Omcirkel het aantal sterren dat het beste past bij wat jij denkt.

Betekenis van de sterren

★ = helemaal niet eens

★★ = niet eens

★★★ = eens

★★★★ = helemaal eens

Vragen

1. Ik vond het leuk om met RekenRangers te werken.

★ ★★ ★★★ ★★★★

2. Ik zou RekenRangers niet nog eens willen gebruiken

★ ★★ ★★★ ★★★★

3. Ik begreep goed wat ik moest doen.



4. Ik had hulp nodig om met het programma te werken.



5. RekenRangers maakte oefenen interessanter



6. Ik begreep goed of mijn antwoord juist of fout was.



7. De oefeningen pasten niet goed bij wat ik kon.



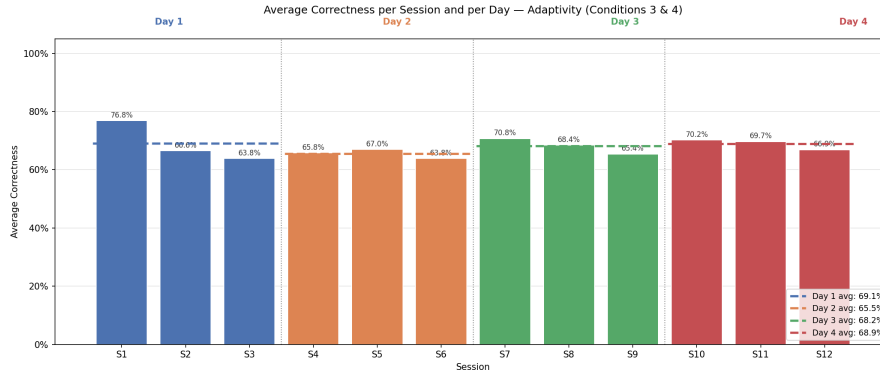
Dankjewel om dit in te vullen!

Appendix F

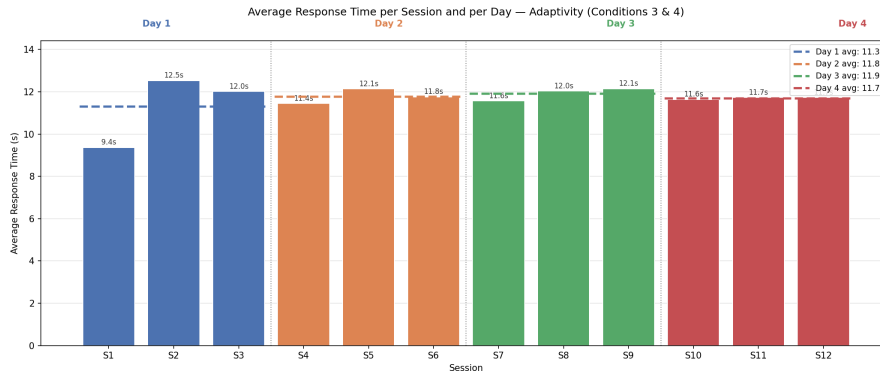
Within-day performance in the adaptive conditions

Section 7.3.2 presents the per-block correctness and response time for the non-adaptive conditions (A and B). For completeness, Figure F.1 shows the equivalent data for the adaptive conditions (C and D).

F. WITHIN-DAY PERFORMANCE IN THE ADAPTIVE CONDITIONS



(A) Average correctness per block.



(B) Average response time per block.

FIGURE F.1: Average correctness and response time per block for the adaptive conditions (C and D), across the twelve blocks of the four training days.

Code of conduct and transparency statement on the use of GenAI for KU Leuven-students (academic year 2025–2026)

Generative AI (GenAI) assistance tools can be used to generate various types of content, including text, images, code, video, music, or combinations thereof. Common examples of such tools include ChatGPT, Google Gemini, Microsoft Copilot, Midjourney, Claude.ai, Perplexity.ai, and DALL · E, among others.

This code of conduct is a tool that helps students to be transparent about the use of GenAI and fits within the university's principles on academic integrity.

Important guidelines and remarks

- **Sensitive or personal data:** Some GenAI tools protect your input and sensitive or personal data better than others. There is often no transparency on what the owners of the AI applications do with the data entered. Therefore, **do not enter sensitive or personal data in free GenAI tools**. More info about what sensitive and personal data are, is described in the [three confidentiality levels of the KU Leuven data classification model](#) (non-confidential, confidential, strictly confidential). For confidential data you should preferably use **M365 Copilot Chat with your KU Leuven account**. If you use another GenAI tool, you must be **absolutely certain** it does **not store or reuse the data you enter**. **Try to avoid entering these data. Do so only if strictly necessary, and only to the extent required. For personal data, work with anonymous or pseudonymized data.** Additionally, in case of strictly confidential data this should first be discussed with the teaching staff of the course or your thesis supervisor.
- **Copyrighted materials:** For lawfully obtained copyrighted material you should preferably use **M365 Copilot Chat with your KU Leuven**

account. If you use another GenAI tool, you must be **absolutely certain** it does **not store or reuse the data you enter**.

- GenAI assistance may not be used for data or topics covered by a **Non-Disclosure Agreement (NDA)**. Even in the absence of an NDA, certain information may still need to be treated as confidential—for example, due to regulatory requirements or the risk of significant harm to the university if disclosed. In case of doubt, check with your teaching staff or supervisor.
- If your master’s thesis is under **embargo**, you should first discuss with your supervisor whether the use of GenAI is permitted.
- Before using a GenAI tool, always consider whether its use is responsible, including from a **sustainability perspective** (e.g. when using GenAI as a search engine, language assistant,...).
- **Take a scientific and critical attitude** when interacting with GenAI assistance and interpreting its output, that may not always be correct.
- As a student you are responsible for complying with Article 84 of the Regulations on Education and Examinations: your report or thesis should reflect your own knowledge, understanding and skills. Be aware that plagiarism rules also apply to (work that is the result of) the use of GenAI assistance tools.
Exam Regulations Article 84: *“Every conduct individual students display with which they (partially) inhibit or attempt to inhibit a correct judgement of their own knowledge, understanding and/or skills or those of other students, is considered an irregularity which may result in a suitable penalty. A special type of irregularity is plagiarism, i.e. copying the work (ideas, texts, structures, designs, images, plans, codes , ...) of others or prior personal work in an exact or slightly modified way without adequately acknowledging the sources. Every possession of prohibited resources during an examination (see article 65) is considered an irregularity.”*
- In order to maintain academic integrity and avoid plagiarism, **more information about being transparent on the use of GenAI assistance and about correctly citing and referencing GenAI** can be found on this website for students (**Dutch/English**).
- **Additional reading:** KU Leuven guidelines on responsible use of Generative AI tools, and other information (**Dutch/English**)

A few final words

If you are uncertain whether or not you should declare your use of GenAI tools, we suggest that you discuss this with your instructor or supervisor. It is always safer to declare GenAI use, even when it is not strictly required. However, declaring GenAI use does not entail that its use is

allowed; the right column in the table below provides more detailed instructions in this regard (code of conduct).

Moreover, advanced AI tools are evolving rapidly, and their capabilities have expanded significantly in a short period of time. As a result, we do not yet have all the answers about their responsible use. Finally, it is important to follow-up on the most recent evolutions in AI technologies, to have an open mind but also to be a bit cautious, to communicate with instructors, teaching assistants, supervisors and peers, to be as transparent as we can, and to learn together as we move along.

Student name: Naten Kindekens **Student number:** R0899275

Student name: Naten Kindekens **Student number:** R0899275

Please indicate with "X" whether it relates to a course assignment or to the Bachelor's or Master's thesis:

O This form is related to a **course assignment**.
Course name: **Course number:**

Course name: Course number:

This form is related to my **O Bachelor's** or **X Master's thesis**.
Title Master's thesis: RekenRangers: A configurable arithmetic learning tool designed for classroom use and educational research
Supervisor: Prof.dr. Tom Schrijvers

Title Master's thesis:	RekenRangers: A configurable arithmetic learning tool designed for classroom use and educational research	Supervisor:	Prof.dr. Tom Schrijvers
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Daily supervisor: Prof. Dr. Tom Schrijvers, Dr. Elieen Bellon

Please indicate with "X":

X I did use GenAI Assistance. In this case **specify which ones**: Claude laude.ai, Anthropic), ChatGPT (OpenAI) and ElevenLabs.

X I did use GenAI Assistance. In this case **specify which ones**: Claude (Claude.ai, Anthropic), ChatGPT (OpenAI) and ElevenLabs.

GenAI assistance used as/for:	Name of the GenAI tool(s) used If helpful, also describe in which way you were using GenAI related to what is specified as code of conduct.	Code of conduct: For each of the categories below, always take into account the important guidelines and remarks mentioned above (e.g. copyrighted data, sensitive or personal data,...).
As a language assistant for reviewing or improving texts I wrote myself	<i>Claude and ChatGPT. Used to check text I wrote myself for typos and grammar mistakes, and to suggest improvements to sentence structure and wording so the text reads more clearly. The text content and ideas remained my own.</i>	This use is similar to using spelling and grammar check tools. In general, you do not have to refer to such kind of GenAI use in the text. However, be careful: When using GenAI tools on texts you did not write yourself to improve the text, you have to refer to the original source or author, otherwise you are committing plagiarism and thus an irregularity.

As a paraphrasing tool	<i>Claude and ChatGPT. Used occasionally and carefully to paraphrase text I wrote myself. I always reviewed the output critically to make sure it did not introduce information from other sources, so all paraphrased text reflects my own work.</i>	This use is allowed except when it is prohibited by the teacher or the program of study. You may paraphrase your own text or texts by an author other than yourself and take inspiration from what a GenAI tool or another tool suggests (unless it is not allowed). In general, you do not have to refer to such kind of GenAI use or other paraphrasing tools in the text. However, be careful: • If it entails text by an author other than yourself, you are not allowed to include that paraphrased text without reference to the original source or author. Without such reference, you would be committing plagiarism and thus an irregularity.
For translation aid to improve texts I wrote myself or to better understand text from others	<i>Not used.</i>	This use is allowed except when it is prohibited by the teacher or the program of study. It is similar to using translation tools (Google translate, DeepL, ...). In general, you do not have to refer to such kind of GenAI use in the text. However, be careful: • You are not allowed to include that translated text without reference to the original source or author. Without such reference, you would be committing plagiarism and thus an irregularity. • Always check the translated text for correctness and meaning.

As a search engine to get information on a topic or to search for existing research on the topic	<i>Claude and ChatGPT. Used to get an initial overview of topics covered in the thesis. I wrote my own text based on this information and verified it against scientific sources rather than copying it.</i>	This use is similar to e.g. a Google search or checking Wikipedia. If you write your own text based on this information, you do not have to refer to the use of GenAI in the text. You only have to refer to the existing research and references you have checked and used (without such references you would be committing plagiarism and thus committing an irregularity). However, be careful: • Be aware that the output of the GenAI tool cannot be guaranteed as a 100% reliable source of information. The output may not be entirely correct and/or be limited due to the databases it uses. Moreover, knowledge evolves and may change over time; therefore, the database of the GenAI tool may not be up to date. Therefore, verify the information and do not just copy-paste it as you should understand and critically process everything you are writing.
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For literature search	<i>Claude and ChatGPT. Used to help discover potentially relevant papers. Every paper was then located and read manually; no paper or citation was used unless I had checked the original source myself.</i>	This use is comparable to e.g. a Google Scholar search. You do not have to refer to such kind of GenAI use; you only have to refer to the literature references you have checked and used (without such references you would be committing plagiarism and thus committing an irregularity). However, be careful: • Be aware that the search output is restricted to the database the GenAI tool is built on. After this initial search, look for scientific sources and conduct your own analysis of the source documents. Interpret, analyse and process the information you obtained; verify it and do not just copy-paste it as you should understand and critically process everything you are writing. • Be aware that some GenAI tools may output no or wrong references. As a student you are responsible for further checking and verifying the absence or correctness of references; do not just copy-paste it.
For generating programming code	<i>Claude and ChatGPT. Used to assist with writing code for the RekenRangers application and to help detect and fix bugs. I reviewed, tested and understood all code that was used.</i>	Use of GenAI for coding is allowed except when it is prohibited by the teacher or the program of study. If used for coding, correctly mention the use of GenAI assistance in accordance with the instructions on the page on being transparant about the use of GenAI.

For generating new ideas (research)	<i>Claude and ChatGPT. Used as a supporting aid to generate ideas when I was stuck. Not all suggestions were used; ideas were critically evaluated and only adopted where appropriate.</i>	This use of GenAI is allowed except when it is prohibited by the teacher or the program of study. Correctly mention the use of GenAI assistance in accordance with the instructions on the page on being transparent about the use of GenAI. Be careful: • Further verify in this case whether the idea is novel or not. It is likely that it is related to existing work. If so, that existing work should be correctly referenced in the text (without such reference you would be committing plagiarism and thus committing an irregularity).
For generating synthetic data	<i>Not used.</i>	Use of GenAI for generating synthetic data is allowed, provided that it is methodologically and ethically justifiable, except when it is prohibited by the teacher or the program of study. Always correctly mention the use of GenAI assistance in accordance with the instructions on the page on being transparent about the use of GenAI. Be careful: • Always carefully evaluate the generated synthetic data for quality and possible bias since the output is highly dependent on the quality of the data on which the models are trained.

For generating blocks of text (other than the allowed use without referencing mentioned above)	<i>Not used.</i>	According to Article 84 of the Regulations on Education and Examinations your text should allow to correctly and properly assess your own knowledge, understanding and skills. Therefore, inserting blocks of text without quotes and a reference to GenAI assistance in your work is not allowed. Be careful: • If it is really needed to insert a block of text from a GenAI tool, for instance because of the nature of your assignment, mention it as a citation by using quotes and correctly mention the use of GenAI assistance in accordance with the instructions page on being transparent about the use of GenAI. • However, in general, such GenAI use should be kept to an absolute minimum; you should always check the original sources.
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For generating visuals, video or audio	<i>Claude and ChatGPT were used to generate images for the RekenRangers application (the safari-themed visual content used in the tool). ElevenLabs was used to generate a speaking voice for the interactive demo of RekenRangers.</i>	Use of GenAI for generating visuals, audio or video is allowed except when it is prohibited by the teacher or the program of study. Be careful: • If used, refer to GenAI for visuals according to the usual referencing style, following the instructions on the page on referencing GenAI. • If you work with existing visuals, audio or video, that existing work should be correctly referenced in the text (without such reference you would be committing plagiarism and thus an irregularity). • Explain the usage in the methods section (if there is one) and optionally attach (or link to) the prompts with the full output (or history).
Other use (specify here; this may also include a combination of types of use mentioned above):	<i>Claude and ChatGPT. Used for advice on the layout of the thesis presentation and poster, i.e. textual suggestions on what to place where. The actual presentation and poster were designed and produced by me.</i>	To make sure other use of GenAI is allowed within the course or thesis, and if so, the conditions that may apply, contact the teaching staff of the course or the supervisor of the thesis beforehand and explain the intended GenAI use. Also inform the programme director. Motivate how you would comply with Article 84 of the exam regulations. Explain the use and the added value of the AI tool you consider to use and how it is in accordance with the assignment or thesis and the KU Leuven guidelines on responsible use of Generative AI tools. Depending on the kind of GenAI use, it may be needed to properly reference it in the text, in accordance with the instructions page on being transparent about the use of GenAI.